

Prefatory Clarification on the Variational Section

This model intentionally adopts a quasi-solution $\phi(r)$, rather than an exact solution. The reason is that an exact solution would eliminate the physical significance of the characteristic scale $r = R$, thereby preventing the reproduction of the essential behavior: scale-responsive structure of space.

If an exact solution were enforced, the scale $r = R$ would likely be absorbed into coordinate redefinitions or symmetry-induced degeneracies. In such a case, R would lose its status as a physically anchored scale and instead become a redundant parameter, thereby obscuring any meaningful scale-dependent structure in the resulting dynamics.

To verify the validity of this quasi-solution, we conducted a variational residual analysis. The results show that the violation of the extremum condition is localized, well controlled, and appears as a physically meaningful structure. This behavior is also visually illustrated in Fig. 1.

The quasi-solution and the variational residual do not signify compromise or defect in any sense. Rather, they constitute the very foundation of the structure developed here and serve as the key to realizing scale-dependent spatial dynamics.

What is important is that this quasi-solution is not adopted as an approximation, but is introduced for its structural role in deriving the effective Λ term, $\Lambda_{\text{eff}}(r)$.

It does not imply that the entire structure of the universe or the fundamental field equations themselves are quasi-solutions.

The aim of this study is to examine whether such a quasi-solution structure can function effectively within a universe that is otherwise described by exact solutions.

For clarity, the term “logarithmic-type” used in this paper does not refer to a strict logarithmic function itself, but rather to a qualitative feature indicating that the functional form exhibits a logarithmic behavior.

Repulsive Gravity Theory

Based on Scalar Fields and the Effective Λ

Misaki Kasai

May 1, 2026

Abstract

In this study, we propose an extension of the Einstein field equations by introducing a repulsive gravity framework based on the scale-responsive geometric repulsion effect of space, derived from a scalar field. This effect is induced by the geometric configuration of the scalar field, from which a scale-dependent effective cosmological term, Λ_{eff} , emerges. The impact of Λ_{eff} depends on spatial scale, potentially redefining the structural properties of gravity, especially at galactic scales and within black hole (BH) interiors. We demonstrate that two key observational and theoretical challenges—the flattening of galactic rotation curves in their outer regions and the avoidance of black hole singularities—can be addressed in a unified and quantitative manner through the common scalar-field-derived term Λ_{eff} .

1 Overview: Galactic Rotation and BH Singularities

Recent astronomical observations have revealed that the rotational velocities of stars in the outer regions of galaxies significantly exceed those predicted by general relativity, even when the observed amount of baryonic matter is taken into account. This discrepancy is known as the *galactic rotation curve problem* and has long been considered difficult to explain without invoking the existence of dark matter [1, 2].

On the other hand, gravitational collapse inside black holes is known, in the classical framework, to lead to a singularity—i.e., a region of infinite curvature—in the limit $r \rightarrow 0$. This phenomenon is considered to signal the breakdown or limitation of general relativity itself [3]. The so-called *singularity problem* strongly suggests the necessity of quantum corrections or structural modifications within gravitational theory.

2 Successes and Limitations of the Λ CDM Model

The Λ CDM model, which serves as the standard framework in modern cosmology, has achieved remarkable success in explaining observational data such as the cosmic microwave background

(CMB), baryon acoustic oscillations (BAO), and Type Ia supernovae with high precision [4]. However, the cosmological constant Λ is defined as a static and uniform quantity, without incorporating any spatial structure or scale dependence.

Moreover, although Λ CDM relies on invisible components such as dark matter and dark energy, the true nature of these entities remains unknown. They function as auxiliary terms in the theoretical framework, and there has been persistent criticism that they lack a solid basis for physical reality [5].

3 Foundational Standpoint

This study attempts to construct a gravitational theory in which **repulsive response is the subject**, in contrast to general relativity, which treats “gravity as attraction (a manifestation of spacetime curvature)” as its primary perspective. Specifically, we propose a framework that introduces an effective cosmological constant term, $\Lambda_{\text{eff}}(r)$, derived from a scalar field. This enables a unified description of gravitational structures ranging from galactic scales to black hole singularity scales. Moreover, in this study, the geometric repulsive response is constructed on the basis of a static, spherically symmetric scalar-field Lagrangian, with the aim of later extending the model to cosmology by regarding the geometric scale ratio $\frac{r}{R}$ as the basis of time evolution in the background universe.

Thus, in the present study, from the standpoint of adopting the one-dimensional geometrical repulsive response structure as the first principle, four-dimensional spacetime and matter are regarded as secondary entities generated from that structure, while covariance is positioned as a means for describing them locally. Within this framework, the cosmological constant Λ is retained in the Einstein field equations as an approximately constant background scale, thereby preserving the philosophical foundation of the Λ CDM model while overcoming its structural limitations.

4 The Scalar Field and the Effective Λ

In this section, we demonstrate that the geometric effective Λ term

$$\Lambda_{\text{eff}}(r) = \frac{4r^2}{(R^2 + r^2)^2}$$

is structurally derived from the variational structure of a spherically symmetric and static scalar field configuration.

Here, R represents the characteristic length scale of each physical regime, while both R and r are treated as dimensionless quantities until calibrated against observations.

The central principle of this study is that the essential repulsive response structure is encoded in a radial one-dimensional sector with characteristic scale R .

Accordingly, we adopt the following basic radial one-dimensional scalar-field Lagrangian for a spher-

ically symmetric and static configuration:

$$\mathcal{L}_\phi = -\frac{1}{2} \left(\frac{d\phi}{dr} \right)^2 - V(\phi(r)),$$

where $V(\phi(r))$ is the scalar potential.

This radial one-dimensional formulation serves to generate a scale-dependent repulsive response in a spherically symmetric and static configuration.

In this study, the fundamental radial variable is not r itself, but the dimensionless scale ratio

$$x \equiv \frac{r}{R},$$

where R is the characteristic scale defined above. To require a regular response structure without singularities on the real axis, we impose the following conjugate pair in the complex x -plane, symmetric with respect to the real x -axis:

$$\zeta_\pm = x \pm i, \quad x \geq 0.$$

By Euler's formula, these can be expressed in polar form as

$$\zeta_\pm = \sqrt{1+x^2} e^{\pm i\theta}, \quad \theta = \arctan\left(\frac{1}{x}\right) \quad (x > 0), \quad \theta(0) = \frac{\pi}{2}.$$

We then define the background profile as the real amplitude response obtained after phase cancellation:

$$\mathcal{B}(x) \equiv \ln \zeta_+ + \ln \zeta_-.$$

Substituting the conjugate polar forms into this definition, we obtain

$$\mathcal{B}(x) = \left[\ln \sqrt{1+x^2} + i\theta \right] + \left[\ln \sqrt{1+x^2} - i\theta \right] = 2 \ln \sqrt{1+x^2} = \ln(1+x^2).$$

Therefore, the logarithmic background profile used in this theory is obtained as the real amplitude response generated through the phase cancellation of the complex-conjugate scale pair. Accordingly, the physical meaning of $\mathcal{B}(x)$ is that it represents a dimensionless geometrical background response that space exhibits with respect to the scale ratio $x = r/R$.

At the characteristic point $x = 1$, namely $r = R$, we have

$$\zeta_\pm = 1 \pm i,$$

and hence

$$\arg \zeta_\pm = \pm \frac{\pi}{4}.$$

Since the phase difference of the complex-conjugate pair is generally $\Delta\theta = 2\theta$, it becomes

$$\Delta\theta = \frac{\pi}{2}$$

at $x = 1$. At this point, the real component x and the magnitude of the imaginary component 1 are balanced. Thus, the characteristic scale R appears as a geometrical balance point in the complex

x -plane. This interpretation clarifies why the characteristic scale $r = R$ must remain explicitly encoded in the scalar-field structure.

With this background profile, the scalar field serving as a *shape function* for the spatial profile of the scale-dependent geometric repulsive response is defined by its radial derivative:

$$\phi(r) = \frac{d\mathcal{B}}{dx} \frac{dx}{dr} = \frac{d}{dr} \ln \left(1 + \frac{r^2}{R^2} \right) = \frac{2r}{R^2 + r^2}.$$

Then, $\phi(r)^2 = \frac{4}{R^2} \cdot \frac{x^2}{(1+x^2)^2}$, $\phi(r)^4 = \frac{4}{R^2} \cdot \frac{4}{R^2} \cdot \left(\frac{x^2}{(1+x^2)^2} \right)^2$, and therefore, in the standard $-\phi^2 + \phi^4$ -type potential that guarantees the symmetry $\phi \leftrightarrow -\phi$, identifying the ϕ^4 term as a higher-order term belonging to the same R -hierarchy (with a single overall prefactor of $\frac{4}{R^2}$) as the ϕ^2 term fixes the coefficient to $\frac{1}{4}R^2$. Accordingly, the potential corresponding to the scalar field $\phi(r)$ takes the form

$$\begin{aligned} V(\phi(r)) &= -\phi^2 + \frac{1}{4}R^2\phi^4 = -\left(\frac{2r}{R^2 + r^2} \right)^2 + \frac{1}{4}R^2 \left(\frac{2r}{R^2 + r^2} \right)^4 \\ &= \underbrace{-\frac{4r^2}{(R^2 + r^2)^2}}_{\text{Leading term}} + \underbrace{\frac{4R^2r^4}{(R^2 + r^2)^4}}_{\text{Second-order correction term}}. \end{aligned}$$

Here, the second term serves as a correction that slightly modifies the primary structure, but the leading term clearly satisfies the relation:

$$V(\phi(r)) \propto -\Lambda_{\text{eff}}(r), \quad \Lambda_{\text{eff}}(r) \equiv \frac{4r^2}{(R^2 + r^2)^2}.$$

In particular, in the regime $r \ll R$, the correction term is significantly smaller than the leading term. In the opposite limit $r \gg R$, their asymptotic behaviors are

$$\underbrace{\frac{r^2}{(R^2 + r^2)^2}}_{\text{Leading term}} \sim \frac{1}{r^2}, \quad \underbrace{\frac{r^4}{(R^2 + r^2)^4}}_{\text{Second-order correction term}} \sim \frac{1}{r^4},$$

indicating that the contribution from the correction term decays rapidly. Furthermore, even at the intermediate scale $r = R$, we find:

$$\underbrace{\frac{4R^2}{(2R^2)^2}}_{\text{Leading term}} = \frac{1}{R^2}, \quad \underbrace{\frac{4R^2R^4}{(2R^2)^4}}_{\text{Second-order correction term}} = \frac{1}{4R^2}.$$

showing that the correction term remains only one-fourth of the leading term.

Therefore, across all spatial scales, the leading term dominates, and it is demonstrated that $\Lambda_{\text{eff}}(r)$ can be directly and naturally derived from the scalar field potential $V(\phi(r))$.

5 Variational Evaluation of the Scalar Field

In this section, we examine whether the Lagrangian of the scalar field $\phi(r)$ —which gives rise to the geometric effective cosmological term $\Lambda_{\text{eff}}(r)$ derived in Section 4—satisfies the Euler–Lagrange extremum condition. In particular, we expect that the extremum condition is locally violated around the characteristic spatial scale $r = R$, and that this violation rapidly diminishes for $r \gg R$. Such behavior is naturally anticipated from the structure of the adopted $\phi(r)$, as it corresponds both to the notion that cosmic acceleration may be closely related to a local violation of the extremum condition and serves as a confirmation that the characteristic scale R is not lost within the structure.

Note that the violation of the extremum condition discussed here does not directly imply a violation of the conservation law $\nabla^\mu T_{\mu\nu} = 0$.

From Section 4, the scalar field and its corresponding potential take the form

$$\phi(r) = \frac{2r}{R^2 + r^2}, \quad V(\phi(r)) = -\phi^2 + \frac{1}{4}R^2\phi^4.$$

For a spherically symmetric and static configuration, the Lagrangian is given by:

$$\mathcal{L}_\phi = -\frac{1}{2} \left(\frac{d\phi}{dr} \right)^2 - V(\phi(r)) = -\frac{1}{2} \left(\frac{d\phi}{dr} \right)^2 + \phi^2 - \frac{1}{4}R^2\phi^4.$$

Thus, the Euler–Lagrange equation derived from this Lagrangian is

$$\frac{d^2\phi}{dr^2} = R^2\phi^3 - 2\phi.$$

Therefore, the deviation from the extremum condition (the variational residual) is defined as follows:

$$\delta(r) \equiv \frac{d^2\phi}{dr^2} - (R^2\phi^3 - 2\phi).$$

By substituting the adopted scalar field $\phi(r)$ into this expression, the variational residual is explicitly given by:

$$\delta(r) = \left[\frac{4r(r^2 - 3R^2)}{(R^2 + r^2)^3} \right] - \left[\frac{8R^2r^3}{(R^2 + r^2)^3} - \frac{4r}{R^2 + r^2} \right].$$

This can be rearranged as:

$$\delta(r) = \frac{4r(r^4 + r^2 + R^4 - 3R^2)}{(R^2 + r^2)^3}.$$

Consequently, the characteristic scale at which the linear term in the numerator vanishes near the origin is $R = \sqrt{3}$. At this scale, the variational residual $\delta(r)$ for the scalar field $\phi(r)$ is numerically evaluated, and the resulting profile is shown in Fig. 1.

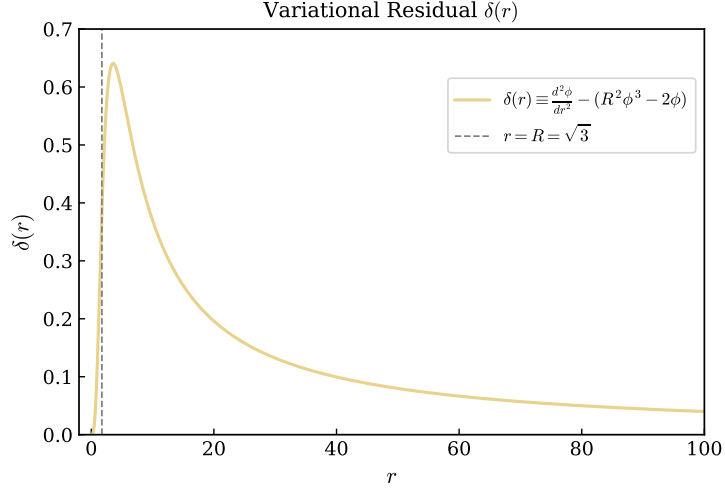


Fig. 1: Variational residual $\delta(r)$ for the scalar field $\phi(r)$.

From Fig. 1, it can be seen that the variational residual $\delta(r)$ begins to locally deviate at the spatial scale $r = R$, reaches a peak shortly thereafter ($r > R$), and then rapidly decays. This result indicates that the scalar field specifically recognizes and responds to $r = R$, and that the violation of the extremum condition is local and non-instantaneous. Such behavior is also consistent with the nature of the sustained cosmic acceleration.

The analysis is performed over the range $r \in [0.0001, 100]$. Within this interval, the maximum value of the variational residual $\delta(r)$ is found to be:

$$\delta_{\max} \approx 0.641 \quad \text{at} \quad r \approx 3.56.$$

To evaluate the significance of this residual, we define the relative error with respect to the extremum condition as:

$$\epsilon_{\phi}(r) \equiv \left| \frac{\delta(r)}{\phi(r)} \right|.$$

At the point where the residual peaks ($r \approx 3.56$), the scalar field takes the value $\phi(r) \approx 0.454$, yielding:

$$\epsilon_{\phi} \approx 1.41.$$

Similarly, at the characteristic scale $r = R = \sqrt{3}$, the relative error is:

$$\epsilon_{\phi}(R) = \left| \frac{\delta(R)}{\phi(R)} \right| \approx 0.667 \quad (\delta(R) \approx 0.385, \phi(R) \approx 0.577).$$

Even around the characteristic scale $r = R$, the relative error remains within a range that can be regarded as a local violation of the extremum condition. The value $\epsilon_{\phi}(R) \approx 0.667$ supports the validity of the function as a quasi-solution.

On the other hand, the variational residual $\delta(r)$ reaches its maximum around $r \approx 3.56$, where the relative error rises to approximately $\epsilon_{\phi} \approx 1.41$. However, since $\phi(r)$ becomes small at this scale, the relative error is numerically overestimated due to the small denominator. Therefore, it is more appropriate to evaluate the validity of the quasi-solution primarily around $r = R$.

Based on these considerations, the scalar field $\phi(r)$ is formulated as a quasi-solution so as to

preserve the characteristic scale R —encoded in the radial one-dimensional structure—rather than as an exact solution that would eliminate it.

In this context, the local violation of the variational extremum condition serves as a necessary condition for the excitation of the scalar field $\phi(r)$.

In this paper, we refer to the structure based on the effective cosmological term $\Lambda_{\text{eff}}(r)$, constructed from the quasi-solution $\phi(r)$, as the Λ_{eff} *model* (or the *f Λ theory*).

6 Consistency with the Observed Λ

The derived scale-dependent effective cosmological term,

$$\Lambda_{\text{eff}}(r) = \frac{4r^2}{(R^2 + r^2)^2},$$

attains its maximum at $r = R$, with the value given by

$$\Lambda_{\text{eff}}^{\text{max}} = \Lambda_{\text{eff}}(r)|_{r=R} = \frac{4R^2}{(R^2 + R^2)^2} = \frac{4R^2}{4R^4} = \frac{1}{R^2}.$$

If this maximum is identified with the observed cosmological constant,

$$\Lambda \approx 10^{-52} \text{ m}^{-2},$$

then it follows that

$$R = \frac{1}{\sqrt{\Lambda}} \approx 10^{26} \text{ m},$$

which coincides with the cosmological scale, $\sim 10^{26} \text{ m}$. Therefore, it is natural to identify this characteristic scale as the effective cosmological scale, $R = \frac{1}{\sqrt{\Lambda}} \equiv R_c$.

This result implies that the effective cosmological term Λ_{eff} —derived from the scalar-field structure proposed in this study—is not only naturally consistent with the observed value of the cosmological constant but also provides a physically consistent origin for cosmic repulsion.

7 Extended Einstein Field Equation

The extended Einstein field equation is written as:

$$\Lambda_{\text{eff}}(r)g_{\mu\nu} + G_{\mu\nu} = \kappa T_{\mu\nu}^{(\text{matter})}, \quad \kappa \equiv \frac{8\pi G}{c^4}.$$

In the local regime where $r \ll R_c$ (e.g., at the solar-system scale), the geometrical term $\Lambda_{\text{eff}}(r)g_{\mu\nu}$ becomes negligibly small, and the field equation naturally reduces to the standard Einstein form:

$$G_{\mu\nu} = \kappa T_{\mu\nu}^{(\text{matter})}.$$

Therefore, this extended field equation remains consistent with well-established observations precisely explained by general relativity, such as the perihelion precession of Mercury and the gravitational redshift.

Here, the effective term $\Lambda_{\text{eff}}(r) = \frac{4r^2}{(R^2 + r^2)^2}$ can be rewritten as:

$$\Lambda_{\text{eff}}(r) = \frac{1}{R^2} F(x), \quad F(x) = \frac{4x^2}{(1+x^2)^2}, \quad x = \frac{r}{R}.$$

By applying the Bianchi identity to the extended Einstein field equation, the exchange current J —representing geometry–matter energy transfer (i.e., local non-conservation)—is defined as follows, and in a static, spherically symmetric configuration, only the radial component remains:

$$J_\nu \equiv \nabla^\mu T_{\mu\nu}^{(\text{matter})} = \frac{1}{\kappa} \partial_\nu \Lambda_{\text{eff}}, \quad J_r = \frac{1}{\kappa R^3} \frac{dF}{dx}.$$

Although local conservation is violated, both the radial and total fluxes vanish:

$$\int_0^\infty J_r dr = 0, \quad \Phi(r) = 4\pi r^2 J_r \Rightarrow \Phi(0) = \Phi(\infty) = 0.$$

Thus, the geometry–matter energy transfer is exactly balanced throughout the volume, ensuring global energy conservation.

Additionally, at cosmological scales $r \sim R_c \sim 10^{26}$ m, where the local non-conservation represented by J_r becomes negligibly small,

$$\left| \frac{d\Lambda_{\text{eff}}}{dr} \right| \sim 10^{-78} \text{ m}^{-3} \Rightarrow \Lambda_{\text{eff}} \approx \text{const.},$$

and hence the Friedmann equations are naturally recovered:

$$H^2 \equiv \left(\frac{\dot{a}}{a} \right)^2 = \frac{8\pi G}{3} \rho - \frac{kc^2}{a^2} + \frac{\Lambda c^2}{3}, \quad \frac{\ddot{a}}{a} = -\frac{4\pi G}{3} \left(\rho + \frac{3P}{c^2} \right) + \frac{\Lambda c^2}{3}.$$

8 Application to Galactic Rotation Curves

8.1 Gravitational Term: Realistic Linear Approximation

In the central region of a galaxy, the baryonic matter is relatively concentrated, allowing the use of a spherically symmetric and uniform density approximation. Under this assumption, the mass $M(r)$ enclosed within a radius r is given by:

$$\rho = \text{const.} \Rightarrow M(r) = \frac{4}{3}\pi r^3 \rho.$$

From the gravitational potential generated by this mass distribution, the rotational velocity is derived as:

$$v_{\text{grav}}^2(r) = \frac{GM(r)}{r} = \frac{G}{r} \left(\frac{4}{3}\pi r^3 \rho \right) = \frac{4}{3}\pi G \rho r^2, \Rightarrow v_{\text{grav}} \propto r.$$

The relation $v_{\text{grav}} \propto r$ agrees with the observed linear rise in the inner regions of galaxies.

8.2 Repulsive Term: Rotational Velocity and Flattening

While the gravitational term from the uniform density model dominates in the central region of galaxies, explaining the flattened rotation curves in the outer regions requires a different dynamical contribution.

In this context, a scale-dependent effective Λ term, generated by the spatial configuration of the scalar field $\phi(r)$, yields a geometric repulsive effect that contributes to the squared rotational velocity as:

$$v_{\text{rep}}^2(r) = v_0^2 \cdot \frac{2r^2}{R^2 + r^2},$$

where:

- v_0 is a characteristic velocity scale (unit: m/s) associated with the repulsive term,
- which becomes dominant in the outer galactic region ($r > R$).

Since this function contributes negligibly near the center and asymptotically approaches a constant value $v_{\text{rep}} \rightarrow \sqrt{2} v_0$ as $r \rightarrow \infty$, it naturally reproduces the observed flattening of galactic rotation curves. Based on observational data, we adopt a representative value of $v_0 \approx 200$ km/s.

8.3 Rotational Velocity from Energy Composition

In this model, the gravitational and repulsive contributions to the rotational velocity are regarded as independent energy components. Therefore, the observed rotational velocity is expressed as a quadratic energy sum:

$$v_{\text{total}}^2(r) = v_{\text{grav}}^2(r) + v_{\text{rep}}^2(r) \quad \Rightarrow \quad v_{\text{total}}(r) = \sqrt{v_{\text{grav}}^2(r) + v_{\text{rep}}^2(r)}.$$

This composite structure implies that:

- in the inner region, $v_{\text{grav}} \propto r$ dominates,
- while in the outer region, $v_{\text{rep}} \rightarrow \sqrt{2} v_0$ flattens the rotation curve.

Overall, this framework demonstrates that the essential features of galactic rotation curves can be reproduced without assuming the existence of dark matter, thereby supporting the validity of the effective Λ term, Λ_{eff} .

8.4 Velocity $v_0 \approx 200$ km/s from the Effective Energy of Λ_{eff}

Based on the result of Section 6 and the vacuum-energy density expression in the Λ CDM model, we obtain the *geometric effective energy density*:

$$\rho_{\Lambda_{\text{eff}}}(R_c) = \rho_{\Lambda} = \frac{\Lambda c^2}{8\pi G} \approx 10^{-26} \text{ kg/m}^3.$$

By integrating this geometric effective energy density over a spherically symmetric region of radius $\sim 1000 \text{ kpc} \approx 3.1 \times 10^{22} \text{ m}$, the corresponding geometric effective mass can be obtained. Since $10^{22} \text{ m} \ll 10^{26} \text{ m}$, the effective energy density $\rho_{\Lambda_{\text{eff}}}(R_c)$ can therefore be regarded as nearly constant within this scale range, yielding:

$$M_{\Lambda} \approx \rho_{\Lambda} \cdot \frac{4}{3}\pi r^3 \approx 10^{42} \text{ kg}.$$

If we convert a part of the effective mass energy into kinetic energy, it follows that:

$$E_{\text{rot}} = \frac{1}{2}M_{\Lambda}v_0^2 \quad \Rightarrow \quad v_0 \approx 200 \text{ km/s} \approx \sqrt{\frac{2E_{\text{rot}}}{M_{\Lambda}}}.$$

Therefore, the characteristic velocity $v_0 \approx 200 \text{ km/s}$ can be naturally estimated within the framework of the Λ_{eff} model, and the corresponding energy $E_{\text{rot}} \approx 2 \times 10^{52} \text{ J}$ is consistent with the typical observational energy scale $E_{\text{obs}} \sim 10^{52} \text{ J}$. Considering that the total geometric effective energy is given by $E_{\Lambda} = M_{\Lambda}c^2 \approx 10^{59} \text{ J}$, the magnitude of E_{rot} is sufficiently small in comparison, ensuring energetic consistency.

These results quantitatively support the physical plausibility of the scale-dependent term Λ_{eff} , derived from the scalar field, as an effective cosmological repulsive component.

8.5 Physical Interpretation: Dominant Structures by Region

- **Central region** ($r < R$):

The gravitational term dominates, with $v_{\text{grav}} \propto r$.

- **Intermediate region** ($r \approx R$):

The repulsive term v_{rep} begins to contribute, marking the transition toward velocity flattening.

- **Outer region** ($r > R$):

The repulsive contribution dominates, and the rotation velocity becomes asymptotically flat, reflecting the logarithmic-type structure and scale dependence of Λ_{eff} .

Representative Parameters and Plotting Conditions

- Gravitational constant: $G = 4.302 \times 10^{-6} \text{ kpc} \cdot (\text{km/s})^2 / M_{\odot}$
- Mass density (average over the galaxy): $\rho = 10^6 M_{\odot} / \text{kpc}^3$
- Characteristic scale of the galaxy: $R = 5 \text{ kpc}$
- Characteristic velocity of the galaxy: $v_0 = 200 \text{ km/s}$
- Radial range for calculation: $r = 0.1 \sim 20 \text{ kpc}$

Using the above representative values, the numerically plotted rotation velocity profile at the galactic scale is shown in Fig. 2, based on the previously derived composite formula:

$$v_{\text{total}}(r) = \sqrt{v_{\text{grav}}^2(r) + v_{\text{rep}}^2(r)}.$$

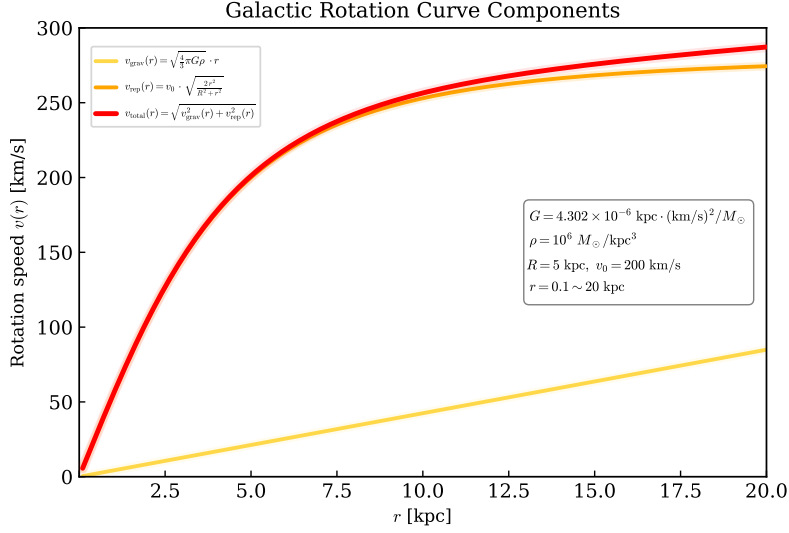


Fig. 2: Numerical simulation of gravitational, repulsive, and total rotational velocity components for a galactic rotation curve.

As clearly demonstrated by the numerical simulation results shown in Fig. 2, it was initially expected that the gravitational term would dominate in the central region ($r < R$). However, the repulsive component was quantitatively found to exceed the gravitational component at all radii within the galaxies modeled. This indicates that the rotational velocity within galaxies is effectively governed by the scalar-field-induced effective Λ term—characterized by its logarithmic-type structure and defined by the characteristic galactic scale $R \sim \text{kpc} \equiv R_g$.

This result not only explains the observed flattening of rotational velocities in the outer regions of galaxies, but also reveals that the repulsive term significantly contributes even in the central region. It strongly indicates that the scale-dominant behavior of the logarithmic-type repulsive term extends far more broadly than previously anticipated.

Therefore, the galactic rotation curves are systematically reproduced by the repulsive component:

$$v_{\text{rep}}(r) = v_0 \cdot \sqrt{\frac{2r^2}{R^2 + r^2}}.$$

9 BH Singularity Avoidance and the Critical Radius

The repulsive acceleration derived from the scalar field $\phi(r)$ is given by:

$$a_{\text{rep}}(r) = c_0 \cdot \phi(r) = \frac{2c_0 r}{R^2 + r^2}.$$

Here, c_0 is a repulsive potential coefficient (unit: m^2/s^2), representing the characteristic spatial energy scale of the repulsive field. Note that this acceleration formula corresponds uniquely to the rotational velocity function defined in Section 8.2 through the circular motion condition $a = \frac{v^2}{r}$:

$$a_{\text{rep}}(r) = \frac{2c_0 r}{R^2 + r^2} \quad \begin{matrix} c_0 = v_0^2 \\ \longleftrightarrow \\ a = \frac{v^2}{r} \end{matrix} \quad v_{\text{rep}}^2(r) = \frac{2v_0^2 r^2}{R^2 + r^2} = v_0^2 \cdot \frac{2r^2}{R^2 + r^2}.$$

Inside the black hole, the repulsive acceleration is considered to balance the gravitational acceleration. Under this condition, the balance between the repulsive and gravitational forces is expressed as:

$$a_{\text{rep}}(r) = g(r) \Rightarrow \frac{2c_0 r}{R^2 + r^2} = \frac{GM}{r^2}.$$

Here, M denotes the mass of the black hole.

At $r = R = r_{\text{crit}}$, where $a_{\text{rep}}(r)$ attains its maximum $a_{\text{rep}}^{\text{max}}$, the balance condition between the repulsive and gravitational accelerations is expressed as:

$$\frac{c_0}{r_{\text{crit}}} = \frac{GM}{r_{\text{crit}}^2} \Rightarrow c_0 = \frac{GM}{r_{\text{crit}}}.$$

Here, the repulsive energy at the critical radius is given by:

$$E_{\text{rep}} = M \cdot c_0 = \frac{GM^2}{r_{\text{crit}}},$$

while the total mass-energy of the black hole is

$$E_{\text{BH}} = Mc^2.$$

Within this setup—where the repulsive response of space is induced by the black hole’s own mass-energy—the system can be treated as a spherically symmetric and static closed configuration. By the principle of energy conservation, the repulsive energy accumulated within the black hole cannot exceed its total mass-energy. Accordingly, equating the two energies defines the critical radius at which further collapse becomes energetically forbidden:

$$E_{\text{rep}} = E_{\text{BH}} \Rightarrow \frac{GM^2}{r_{\text{crit}}} = Mc^2 \Rightarrow r_{\text{crit}} = \frac{GM}{c^2}.$$

This critical radius corresponds exactly to half of the Schwarzschild radius:

$$r_{\text{crit}} = \frac{1}{2} r_s, \quad \text{where} \quad r_s = \frac{2GM}{c^2}.$$

Thus, the characteristic scale of a black hole is naturally defined as $R = \frac{1}{2} r_s \equiv R_{\text{BH}}$.

Since the condition $M \cdot c_0 = Mc^2$ ($E_{\text{rep}} = E_{\text{BH}}$) is satisfied at the critical radius, it follows that $c_0 = c^2$ at that point. **Consequently, singularities are prohibited by the principle of energy conservation and the speed-of-light limit.** The singularity is therefore avoided not by introducing an external cutoff, but by the internal repulsive structure of space itself.

As a direct geometric consequence, the apparent black hole shadow is characterized by the critical radius $r_{\text{crit}} = \frac{GM}{c^2}$. The photon sphere is located at

$$r_{\text{ph}} = \frac{3}{2} r_s = 3 r_{\text{crit}},$$

and the apparent shadow radius, including gravitational lensing, is

$$b_c = \frac{r_{\text{ph}}}{\sqrt{1 - \frac{2r_{\text{crit}}}{r_{\text{ph}}}}} = 3\sqrt{3}r_{\text{crit}}.$$

Hence, the shadow diameter becomes

$$D = 2b_c = 6\sqrt{3}r_{\text{crit}},$$

yielding a dimensionless ratio:

$$\frac{D}{r_{\text{crit}}} = 6\sqrt{3} \approx 10.4,$$

which is universally applicable to any black hole without normalization and agrees remarkably well with the Event Horizon Telescope (EHT) measurements, M87* $d/\theta_g \approx 11.1$ [6] and Sgr A* $d/\theta_g \approx 10.8$ [7].

With $a_{\text{rep}}^{\text{max}}$ the repulsive acceleration at r_{crit} , and since $c_0 = c^2$ there, we have:

$$a_{\text{rep}}^{\text{max}} = \frac{c^2}{r_{\text{crit}}}.$$

Substituting $r_{\text{crit}} = \frac{GM}{c^2}$ gives:

$$a_{\text{rep}}^{\text{max}} = \frac{c^2}{\frac{GM}{c^2}} = \frac{c^4}{GM}.$$

Therefore, both the repulsive acceleration and the gravitational acceleration inside a black hole cannot exceed this value.

Here, we take $a_{\text{rep}}^{\text{max}}$ to be equal to the Planck acceleration $a_{\text{P}} \equiv \sqrt{\frac{c^7}{\hbar G}}$, and we obtain:

$$\frac{c^4}{GM} = \sqrt{\frac{c^7}{\hbar G}} \implies M = \sqrt{\frac{\hbar c}{G}}.$$

This value corresponds to the Planck mass $m_{\text{P}} \equiv \sqrt{\frac{\hbar c}{G}}$, showing that the mass of a black hole, even when compressed to the Planck scale, cannot fall below this fundamental limit.

10 Consistency with the Higgs Mechanism

Following the foundational standpoint of Section 3, matter and four-dimensional spacetime are regarded in the present framework as secondary entities generated from the one-dimensional geometrical repulsive response structure. Therefore, the origin of mass must be identified as a manifestation of the same scalar-field structure, without being introduced as an external postulate. Accordingly, this section clarifies that the scalar potential of the Λ_{eff} model defined in Section 4,

$$V(\phi(r)) = -\phi^2 + \frac{1}{4}R^2\phi^4,$$

is directly embedded in the mass-generating structure of the Higgs mechanism.

To obtain the vacuum expectation value (VEV), we differentiate the potential $V(\phi(r))$ in the Λ_{eff} model:

$$\frac{dV(\phi(r))}{d\phi} = 0 \quad \Rightarrow \quad \phi(-2 + R^2\phi^2) = 0.$$

Therefore, the solutions are

$$\phi = 0 \quad \text{or} \quad \phi = \pm \frac{\sqrt{2}}{R}.$$

Thus, excluding the trivial solution $\phi = 0$, the vacuum expectation value in the Λ_{eff} model is

$$v = \frac{\sqrt{2}}{R}, \quad \phi = \pm v.$$

From this relation, the characteristic scale R can be expressed in terms of the experimentally determined Higgs vacuum expectation value $v \approx 246 \text{ GeV}$ as

$$R = \frac{\sqrt{2}}{v} = \frac{\sqrt{2}}{246 \text{ GeV}} \approx 5.75 \times 10^{-3} \text{ GeV}^{-1}.$$

Using the natural unit conversion ($\hbar = c = 1$) with $\text{GeV}^{-1} \approx 1.97 \times 10^{-16} \text{ m}$, we obtain

$$R \approx 5.75 \times 10^{-3} \times 1.97 \times 10^{-16} \text{ m} \approx 1.13 \times 10^{-18} \text{ m}.$$

This value precisely corresponds to the electroweak scale at which the Higgs boson was discovered in experiments at the Large Hadron Collider (LHC), at $\sim 10^{-18} \text{ m}$. Therefore, it is natural to identify this characteristic scale as the Higgs scale, $R = \frac{\sqrt{2}}{v} \equiv R_h$, and the Yukawa coupling can be defined through a geometric relation as

$$y_f = \frac{R_h}{R_f}, \quad R_f \equiv \frac{1}{m_f},$$

or, taking logarithms, as a distance in log-scale space:

$$-\ln y_f = \Delta\nu, \quad \Delta\nu \equiv \ln R_f - \ln R_h.$$

Here, the potential for the Higgs field in the Standard Model is given by:

$$V_{\text{Higgs}}(\phi) = -\frac{1}{2}\mu^2\phi^2 + \frac{1}{4}\lambda\phi^4, \quad v^2 = \frac{\mu^2}{\lambda}, \quad m_h^2 = 2\lambda v^2.$$

By substituting these relations, the potential can be rewritten as:

$$V_{\text{Higgs}}(\phi) = -\frac{1}{4}m_h^2\phi^2 + \frac{m_h^2}{8v^2}\phi^4.$$

Thus, the Higgs potential can be expressed in a subdivided form:

$$V_{\text{Higgs}}(\phi) = -\left(\frac{m_h}{2}\right)^2\phi^2 + \frac{1}{4}\left(\frac{\sqrt{2}}{v}\right)^2\left(\frac{m_h}{2}\right)^2\phi^4.$$

Here, let us define:

$$\frac{m_h}{2} \equiv \check{m},$$

and substitute the previously obtained relation $R_h = \frac{\sqrt{2}}{v}$:

$$V_{\text{Higgs}}(\phi) = -\check{m}^2 \phi^2 + \frac{1}{4} R_h^2 \check{m}^2 \phi^4.$$

Rearranging gives

$$V_{\text{Higgs}}(\phi) = \check{m}^2 \left(-\phi^2 + \frac{1}{4} R_h^2 \phi^4 \right).$$

This suggests the natural choice

$$\check{m} = 1$$

to bridge mass-dimensional and dimensionless forms. By adopting $\hbar = c = \check{m} = 1$ and replacing $R_h = \frac{\sqrt{2}}{v}$ with a general characteristic scale R , we obtain a universal scalar potential in a mass-dimensionless form:

$$V_{\text{Higgs}}(\phi) = -\phi^2 + \frac{1}{4} R^2 \phi^4$$

which can be applied to general scales and is identical to the potential of the Λ_{eff} model $V(\phi(r)) = -\phi^2 + \frac{1}{4} R^2 \phi^4$.

By assigning $\check{m}$ as the characteristic mass in $V(\phi(r))$ at the Higgs scale, the self-coupling constant is obtained as:

$$\lambda_{\Lambda_{\text{eff}}} \equiv R_h^2 \check{m}^2.$$

In natural units ($\hbar = c = 1$), where $\check{m} \neq 1$ is explicitly retained as the mass scale, using $R_h \approx 1.13 \times 10^{-18} \text{ m} \approx 5.75 \times 10^{-3} \text{ GeV}^{-1}$ and $\check{m} = \frac{m_h}{2} \approx 62.5 \text{ GeV}$, we obtain

$$\lambda_{\Lambda_{\text{eff}}} \approx (5.75 \times 10^{-3} \text{ GeV}^{-1})^2 \cdot (62.5 \text{ GeV})^2 \approx 0.129,$$

which coincides with the Standard Model Higgs self-coupling constant, $\lambda_{\text{SM}} \approx 0.129$.

These results in this section indicate that mass generation is driven by the geometric repulsive response.

Is Double-Higgs (hh) Production Possible?

The production of a single Higgs boson (h) occurs when two half-mass quantum states $|m_h/2\rangle_{1,2}$ coherently superpose within the squared region of the characteristic R_h -scale:

$$\frac{1}{\sqrt{2}} \left(|m_h/2\rangle_1 + |m_h/2\rangle_2 \right) \Rightarrow |h\rangle.$$

This coherent-overlap condition is reflected in the relation

$$\lambda_{\Lambda_{\text{eff}}} = R_h^2 \left(\frac{m_h}{2} \right)^2.$$

In contrast, the production of a double-Higgs (hh) consists of two independent single-Higgs superposition states,

$$|hh\rangle = |h\rangle \otimes |h\rangle = \left(\frac{1}{\sqrt{2}}\right)^2 \left(|m_h/2\rangle_1 + |m_h/2\rangle_2\right) \otimes \left(|m_h/2\rangle_3 + |m_h/2\rangle_4\right).$$

However, since the characteristic length scale $R_h \approx 1.13 \times 10^{-18}$ m corresponds to an energy scale $1/R_h \approx 174$ GeV, localizing four half-mass components with total energy ≈ 250 GeV within the same R_h -scale region would exceed this limit:

$$E_{hh} \equiv 4 \cdot \left(\frac{m_h}{2}\right) \approx 250 \text{ GeV} > E_{R_h} \equiv \frac{1}{R_h} \approx 174 \text{ GeV}.$$

Consequently, the four-point overlap — i.e., double-Higgs production — is energetically prohibited by the R_h -scale confinement limit.

11 Discussion and Conclusion

In this study, we derived the following scale-dependent effective cosmological term from the scalar field $\phi(r)$, which is formulated as a quasi-solution that preserves the characteristic scale R while remaining consistent with the variational principle:

$$\Lambda_{\text{eff}}(r) = \frac{4r^2}{(R^2 + r^2)^2}.$$

This term functions effectively in the following three aspects:

1. Consistency with cosmology:

The effective term Λ_{eff} reaches its maximum value at $r = R$, where

$$\Lambda_{\text{eff}}^{\text{max}} = \frac{1}{R^2}.$$

By identifying this with the observed cosmological constant, we obtain $R \approx 10^{26}$ m, which coincides with the cosmological scale, $\sim 10^{26}$ m.

2. Reproduction of galactic rotation curves:

Using the repulsive term based on Λ_{eff} , we constructed the following composite formula for rotational velocity:

$$v_{\text{total}}(r) = \sqrt{v_{\text{grav}}^2(r) + v_{\text{rep}}^2(r)}, \quad v_{\text{grav}}^2(r) = \frac{4}{3}\pi G\rho r^2, \quad v_{\text{rep}}^2(r) = v_0^2 \cdot \frac{2r^2}{R^2 + r^2}.$$

Numerical evaluation under a realistic galactic-scale configuration and representative parameters confirmed that this structure successfully reproduces the observed flat rotation velocity profiles of galaxies. As a result, the geometric repulsive term associated with Λ_{eff} was found to be dominant over the gravitational contribution throughout the galaxy.

Therefore, the galactic rotation curves are systematically characterized by the repulsive component:

$$v_{\text{rep}}(r) = v_0 \cdot \sqrt{\frac{2r^2}{R^2 + r^2}}.$$

3. Critical radius and singularity avoidance

By deriving the condition under which the repulsive energy equals the mass–energy of the black hole, we obtained a critical radius $r_{\text{crit}} = \frac{1}{2}r_s$ at which collapse is doubly locked by energy conservation and the speed-of-light limit. This indicates that singularities can be naturally avoided within the internal dynamics of the geometric structure itself, without requiring any external cutoff.

Consequently, the contribution of Λ_{eff} manifests as a geometrical effect opposite in direction to conventional gravity, playing a unifying role across scales—from the accelerated expansion of the universe and the flattening of galactic rotation curves to the prevention of black hole singularities.

Continued theoretical advancement of the model, along with its observational and empirical verification, remains a central task. As a first step in this direction, we have confirmed its consistency with the Higgs mechanism within the scalar potential framework of the f Λ theory.

Acknowledgements

This research was solely developed by the author, without external collaboration.

Funding Statement

No external funding was received for this work.

Ethical Statement

This study involved no human or animal subjects; ethical approval was not applicable.

Conflict of Interest Statement

The author declares no conflict of interest.

References

- [1] V. C. Rubin and W. K. Ford, Jr. Rotation of the Andromeda Nebula from a Spectroscopic Survey of Emission Regions. *The Astrophysical Journal*, 159:379–403, 1970.
 - [2] Y. Sofue and V. Rubin. Rotation Curves of Spiral Galaxies. *Annual Review of Astronomy and Astrophysics*, 39:137–174, 2001.
 - [3] Roger Penrose. Gravitational Collapse and Space-Time Singularities. *Physical Review Letters*, 14(3):57–59, 1965.
 - [4] Planck Collaboration. Planck 2018 results. VI. Cosmological parameters. *Astronomy & Astrophysics*, 641:A6, 2020.
 - [5] T. Padmanabhan. Cosmological Constant - The Weight of the Vacuum. *Physics Reports*, 380:235–320, 2003.
 - [6] Event Horizon Telescope Collaboration. First M87 Event Horizon Telescope Results. VI. The Shadow and Mass of the Central Black Hole. *The Astrophysical Journal Letters*, 875(1):L6, 2019.
 - [7] Event Horizon Telescope Collaboration. First Sagittarius A* Event Horizon Telescope Results. IV. Variability, Morphology, and Black Hole Mass. *The Astrophysical Journal Letters*, 930(2):L15, 2022.
-

Appendix A

Cosmological Mapping of the Variational Residual Structure

In Section 3, the geometric scale ratio $\frac{r}{R}$ was positioned as the basis of time evolution. On this basis, this appendix summarizes how $\Lambda_{\text{eff}}(r)$ and the variational residual structure in the present theory (f Λ) are mapped onto cosmology.

From Section 5, the variational residual at $R = \sqrt{3}$ is

$$\delta(r) = \frac{4r^3(r^2 + 1)}{(r^2 + 3)^3}$$

and it takes its maximum value $\delta_{\text{max}} \simeq 0.641$ at $r_{\text{peak}} \simeq 3.56$. As noted in Section 4, since both r and R are treated as dimensionless quantities, r_{peak} is also dimensionless.

From Section 7, the base response function of $\Lambda_{\text{eff}}(r)$ is

$$F(x) = \frac{4x^2}{(1 + x^2)^2}, \quad x = \frac{r}{R}.$$

Here, since the physical length in FLRW spacetime is given by $\ell_{\text{phys}}(t) = a(t) \ell_{\text{com}}$, we have $x(t) \equiv \frac{\ell_{\text{phys}}(t)}{R} = \frac{a(t) \ell_{\text{com}}}{R}$. Therefore, if we normalize the comoving scale as $\ell_{\text{com}} = R$ with respect to the characteristic scale R , then $x(t) = a(t)$. Accordingly, with the standard relation $a(z) = (1 + z)^{-1}$, the cosmological mapping is

$$x(z) = a(z) = (1 + z)^{-1}$$

and, by defining $\Xi_{\text{base}}(z) \equiv F(x(z))$, we obtain

$$\Xi_{\text{base}}(z) = \frac{4(1 + z)^2}{[(1 + z)^2 + 1]^2}.$$

If one assumes that the residual strength is reflected in local cosmological expansion, then from the dimensionless relation $\frac{r}{r_{\text{peak}}} = \frac{z}{z_c}$ (i.e., $r = r_{\text{peak}} \cdot \frac{z}{z_c}$), the variational residual $\delta(r)$ is expressed as

$$\delta\left(r_{\text{peak}} \cdot \frac{z}{z_c}\right).$$

Normalizing this by δ_{max} , one obtains the normalized window function

$$W_{\delta}(z; z_c) = \frac{\delta\left(r_{\text{peak}} \cdot \frac{z}{z_c}\right)}{\delta_{\text{max}}}$$

which is directly proportional to the variational residual and satisfies both $W_{\delta}(0; z_c) = 0$ and $W_{\delta}(z_c; z_c) = 1$. Therefore, the effective expansion response function is given by

$$\Xi(z) = \Xi_{\text{base}}(z) [1 + \epsilon W_{\delta}(z; z_c)]$$

and the dimensionless expansion rate and the Hubble parameter are respectively written as

$$E^2(z) = \Omega_r(1+z)^4 + \Omega_m(1+z)^3 + \Omega_\Lambda \Xi(z), \quad H(z) = H_0 E(z).$$

In this study, we assume an approximately flat spacetime background and adopt the representative values $\Omega_m(\equiv \Omega_{m0}) = 0.30$, $\Omega_r = 9 \times 10^{-5}$, $\Omega_\Lambda = 1 - \Omega_m - \Omega_r$, and $H_0 = 67.4 \text{ km s}^{-1} \text{ Mpc}^{-1}$. We also apply the maximum residual value δ_{max} as the normalization coefficient, namely,

$$\epsilon = \delta_{\text{max}} \simeq 0.641.$$

A.1 Effective Redshift Scale and the H_0 Tension

The quantity z_c is the effective redshift scale, and it is natural to regard its value as being determined by the distance to the cosmic microwave background (CMB) at each cosmic epoch. Therefore, to determine the value of z_c today, it is sufficient to evaluate the point at which the relative difference in the comoving distance to the last-scattering surface at $z_* = 1089.9$,

$$\Delta D_C = \frac{D_C^{\text{f}\Lambda} - D_C^{\Lambda\text{CDM}}}{D_C^{\Lambda\text{CDM}}},$$

is minimized.

Table 1: Relative difference in the comoving distance to the last-scattering surface, $\Delta D_C(z_*)$, and its dependence on z_c . The calculation was performed for $z_* = 1089.9$, using $\Omega_m = 0.30$, $\Omega_r = 9 \times 10^{-5}$, $\Omega_\Lambda = 1 - \Omega_m - \Omega_r$, $r_{\text{peak}} = 3.56$, and $\epsilon = \delta_{\text{max}} = 0.641$.

z_c	$\Delta D_C(z_* = 1089.9)$
0.020	+0.762 %
0.050	+0.304 %
0.080	+0.0204 %
0.0820	+0.00515 %
0.0827	−0.000100 %
0.0830	−0.00234 %
0.100	−0.117 %
0.150	−0.349 %
0.200	−0.483 %

From this result, we find that the relative difference in the comoving distance to the last-scattering surface is minimized, while changing sign, in the vicinity of $z_c \simeq 0.0827$. Therefore, the effective redshift scale today is determined to be

$$\boxed{z_c \simeq 0.0827}.$$

Using this value of z_c , we define H_0^{app} as the best-fit value of H_0 obtained by fitting the ΛCDM distance modulus $\mu_{\Lambda\text{CDM}}(z; H_0)$ to the distance modulus of the present model,

$$\mu_{\text{f}\Lambda}(z) = 5 \log_{10} \left(\frac{d_L^{\text{f}\Lambda}(z)}{\text{Mpc}} \right) + 25, \quad d_L^{\text{f}\Lambda}(z) = (1+z) \frac{c}{H_0} \int_0^z \frac{dz'}{E_{\text{f}\Lambda}(z')},$$

evaluated over the Hubble-flow range $0.0233 < z < 0.15$ in the unweighted least-squares sense, which yields

$$H_0^{\text{app}} \simeq 73.6 \text{ km s}^{-1} \text{ Mpc}^{-1}.$$

This means that, while retaining the early-universe Λ CDM reference value $67.4 \text{ km s}^{-1} \text{ Mpc}^{-1}$, a larger apparent Hubble constant is observed in the local universe around $z_c \simeq 0.0827$, which is also consistent with the representative value $H_0 \simeq 73 \text{ km s}^{-1} \text{ Mpc}^{-1}$ in the range $0.0233 < z < 0.15$ reported by SH0ES [8].

A.2 S_8 Tension

To maintain consistency in how the variational residual is reflected in the expansion sector, we introduce the same window function W_δ into the effective gravitational coupling,

$$\mu(z) = 1 - \eta W_\delta(z; z_c),$$

with the normalization coefficient $\eta = \delta_{\text{max}} \simeq 0.641$.

With this effective coupling $\mu(z)$, the linear growth equation is given by

$$\frac{d^2 D}{da^2} + \frac{3 + \frac{d \ln H}{d \ln a}}{a} \frac{dD}{da} - \frac{3}{2a^2} \mu(z) \Omega_m(a) D = 0, \quad \Omega_m(a) = \frac{\Omega_{m0} a^{-3}}{E(a)^2}, \quad a = (1+z)^{-1},$$

which we solve numerically, yielding

$$S_8 = \sigma_8 \sqrt{\frac{\Omega_{m0}}{0.3}} \simeq 0.778.$$

This result implies a smaller amplitude of structure growth than that predicted by standard Λ CDM, and is also consistent with the value $S_8 \simeq 0.77$ inferred from weak-lensing observations such as KiDS [9]. (In this evaluation, the primordial fluctuation amplitude is taken to be the same as in Λ CDM, and σ_8 is estimated from the ratio of the growth function $D(1)$ at $a = 1$, using $\sigma_{8,\Lambda\text{CDM}} = 0.811$ as the reference value [4]. Since we adopt $\Omega_{m0} = 0.30$ in this work, we have $S_8 = \sigma_8$ in this case.)

Therefore, the variational residual structure derived from the present theory simultaneously alleviates the H_0 tension and the S_8 tension identified in standard cosmology, while preserving the geometry of the CMB, as summarized below:

$$\boxed{\begin{array}{l} z_c \simeq 0.0827, \quad \epsilon = \eta = \delta_{\text{max}} \\ H_0^{\text{app}} \simeq 73.6 \text{ km s}^{-1} \text{ Mpc}^{-1}, \quad S_8 \simeq 0.778, \quad \Delta D_C(z_* = 1089.9) \approx -0.000100 \% \end{array}}.$$

Finally, the behavior of the dimensionless expansion rate $E(z)$ in the present model, compared with that of Λ CDM, is shown in Fig. 3.

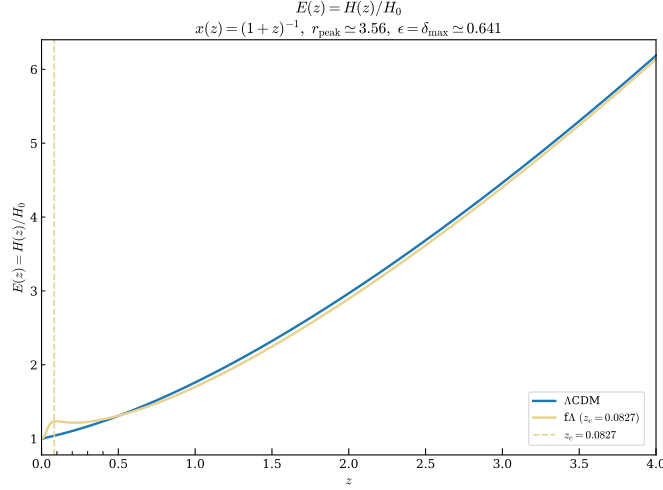


Fig. 3: Behavior of the dimensionless expansion rate $E(z)$ compared with Λ CDM.

In addition, a comparison of $f\sigma_8(z)$ with overlaid observational data points from Table I of [10], approximately corrected to the reference value $\Omega_{m0} = 0.30$, is presented in Fig. 4. Here, as in Λ CDM, $f\sigma_8(z) = f(z)\sigma_8(z)$, where the growth rate is defined by $f(z) \equiv \frac{d \ln D}{d \ln a}$.

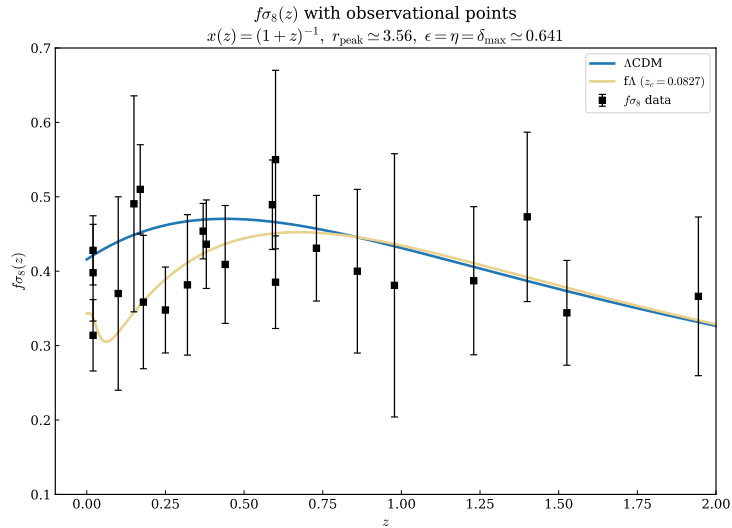


Fig. 4: Comparison of $f\sigma_8(z)$ with observational data points overlaid. The $f\sigma_8(z)$ curve of the present model (yellow) can be seen to reproduce the observed structure-growth data relatively well.

Additional References

- [8] A. G. Riess et al. A Comprehensive Measurement of the Local Value of the Hubble Constant with $1 \text{ km s}^{-1} \text{ Mpc}^{-1}$ Uncertainty from the Hubble Space Telescope and the SH0ES Team. *The Astrophysical Journal Letters*, 934:L7, 2022.
- [9] S.-S. Li et al. KiDS-1000: Cosmology with improved cosmic shear measurements. *Astronomy & Astrophysics*, 679:A133, 2023.
- [10] B. Sagredo, S. Nesseris, and D. Sapone. Internal Robustness of Growth Rate data. *Physical Review D*, 98:083543, 2018, arXiv:1806.10822 [astro-ph.CO].

B The Matter-Like Geometric Component Resisting the Variational Residual and Ω_{m0}

In this appendix, we formulate the critical restoring response that can resist the rise of the variational residual as a matter-like effective geometric component.

As shown in Section 4, the scalar field $\phi(r)$ is treated in the present theory as a dimensionless amplitude characterizing the geometric response. Under the assumption of spherical symmetry, the normalized amplitude of the local geometric response is represented by the dimensionless amplitude $\phi(r)$ normalized by the radius r , namely $\phi(r)/r$.

To describe this amplitude as an effective component in the cosmological background, it is necessary to use the cosmological length scale $R_c = 1/\sqrt{\Lambda}$, defined in Section 6, as the reference length in the denominator. That is, $\phi(r)/R_c$ represents the amplitude of the local geometric response normalized to the cosmological background scale, and its strength is characterized by $(\phi(r)/R_c)^2$. To convert this strength into the form of an effective density, we define the effective density of the geometric resistive component using the gravitational coupling constant $\kappa = 8\pi G/c^4$ as

$$\rho_{\text{geo}}^{\text{resist}}(r) \equiv \frac{1}{\kappa} \left(\frac{\phi(r)}{R_c} \right)^2.$$

Here, as discussed in Appendix A, $x(z) = a(z) = (1+z)^{-1}$ under the comoving scale normalization $\ell_{\text{com}} = R$. Therefore, from the relation $\ell_{\text{phys}} = a\ell_{\text{com}}$ at the present epoch, namely at $a = 1$, it follows that

$$r = \ell_{\text{phys}} = \ell_{\text{com}} = R.$$

Accordingly, the present value of the effective density of the geometric resistive component is given by

$$\rho_{\text{geo},0}^{\text{resist}} = \frac{1}{\kappa} \left(\frac{\phi(R)}{R_c} \right)^2.$$

Normalizing this by the present critical density $\rho_{\text{crit},0} = \frac{3H_0^2 c^2}{8\pi G}$, we obtain

$$\Omega_{\text{geo},0}^{\text{resist}} = \left(\frac{\phi(R)}{R_c} \right)^2 \frac{c^2}{3H_0^2}.$$

Moreover, since Section 6 gives $1/R_c^2 = \Lambda$, this can be reduced to

$$\Omega_{\text{geo},0}^{\text{resist}} = \phi(R)^2 \frac{\Lambda c^2}{3H_0^2} = \phi(R)^2 \Omega_{\Lambda_{\text{eff}},0},$$

where $\Omega_{\Lambda_{\text{eff}},0}$ denotes the cosmological component obtained by normalizing the geometric effective density $1/(R_c^2 \kappa)$ (corresponding to the cosmological-scale base strength $1/R_c^2$ multiplying the base response function $F(x)$ in $\Lambda_{\text{eff}}(r)$ as rewritten in Section 7) by the present critical density $\rho_{\text{crit},0}$, namely $\Omega_{\Lambda_{\text{eff}},0} \equiv (1/R_c^2) c^2 / (3H_0^2) = \Lambda c^2 / (3H_0^2)$. Therefore, the present value of the geometric resistive component is given as a cosmological component weighted by the geometric-response strength $\phi(R)^2$. Furthermore, since $\phi(R)^2 = 1/3$ at the characteristic scale $R = \sqrt{3}$ determined by the structure of the variational residual, we obtain

$$\Omega_{\text{geo},0}^{\text{resist}} = \frac{1}{3} \Omega_{\Lambda_{\text{eff}},0}.$$

That is, the present value of the geometric resistive component has a magnitude corresponding to one third of the cosmological component associated with Λ_{eff} . Moreover, as follows from Sections 4 and 7, Λ_{eff} is treated not as vacuum energy but as a geometric term, and therefore no independent fluid pressure has been postulated. Accordingly, at the background level, this component can be treated, in parallel with Ω_{b0} , as a pressureless matter-like effective component ($w = 0$), and its redshift dependence in the geometric effective density is given, in the standard manner, by

$$\rho_{\text{geo}}^{\text{resist}}(z) = \rho_{\text{geo},0}^{\text{resist}} (1+z)^3.$$

Here, using $x(z) = a(z) = (1+z)^{-1}$, this can be rewritten as

$$\rho_{\text{geo}}^{\text{resist}}(z) = \rho_{\text{geo},0}^{\text{resist}} x(z)^{-3},$$

and furthermore, since $x(z) = \ell_{\text{phys}}(z)/R$,

$$x(z)^{-3} = \left(\frac{R}{\ell_{\text{phys}}(z)} \right)^3.$$

It thus follows that the matter-like redshift dependence of the geometric resistive component is described by this geometric scale ratio. That is, the matter-like geometric resistive component that opposes the rise of the variational residual is characterized at the present epoch by $\phi(R)^2$, in the same manner as $\Lambda_{\text{eff}}(R)$, and in the background universe it rapidly decays, as $\ell_{\text{phys}}(z)$ increases with cosmic expansion, according to $x(z)^{-3}$, namely $\left(\frac{R}{\ell_{\text{phys}}(z)} \right)^3$. Then, the total matter component can be described as the sum of the baryonic matter component and the geometric matter-like component. Therefore, under the approximate flatness condition

$$\Omega_{r0} + \Omega_{b0} + \Omega_{\text{geo},0}^{\text{resist}} + \Omega_{\Lambda_{\text{eff}},0} = 1,$$

we obtain

$$\Omega_{\Lambda_{\text{eff}},0} = \frac{1 - \Omega_{r0} - \Omega_{b0}}{1 + \frac{1}{3}}.$$

Here, since the total matter component is

$$\Omega_{m0} = \Omega_{b0} + \Omega_{\text{geo},0}^{\text{resist}},$$

and since $\Omega_{\text{geo},0}^{\text{resist}} = \frac{1}{3} \Omega_{\Lambda_{\text{eff}},0}$, it follows that

$$\Omega_{m0} = \Omega_{b0} + \frac{1}{3} \cdot \frac{1 - \Omega_{r0} - \Omega_{b0}}{1 + \frac{1}{3}} = \Omega_{b0} + \frac{1 - \Omega_{r0} - \Omega_{b0}}{4}.$$

Therefore, substituting the representative values

$$\Omega_{b0} = 0.049, \quad \Omega_{r0} = 9 \times 10^{-5}$$

into the above expressions, we find quantitatively

$$\boxed{\Omega_{\Lambda_{\text{eff}},0} \simeq 0.71318, \quad \Omega_{\text{geo},0}^{\text{resist}} \simeq 0.23773, \quad \Omega_{m0} \simeq 0.28673}.$$

Using these values and following the same procedure as in Appendix A, we obtain

$$\boxed{z_c \simeq 0.08951}$$

as well as

$$\boxed{H_0^{\text{app}} \simeq 73.3 \text{ km s}^{-1} \text{ Mpc}^{-1}, \quad S_8 \simeq 0.758, \quad \Delta D_C(z_* = 1089.9) \approx +1.91 \times 10^{-6} \%}$$

From the above, it is shown that, under the approximate flatness condition, the geometric matter-like resistive component yields realistic cosmological parameters, and that $\Omega_{\text{geo},0}^{\text{resist}}$ can be interpreted as a matter-like restoring response that suppresses deviations from the variational extremum condition $\delta = 0$. In other words, this restoring response contributes to suppressing the rise of the finite-width, hill-like structure of the variational residual $\delta(r)$, while its present value remains only one third of $\Omega_{\Lambda_{\text{eff}},0}$, so that it cannot become dominant at the boundary $r = R$, where the transition from restoring resistance to geometric repulsive response begins; moreover, once r exceeds R , this resistance rapidly weakens. As a result, the geometric repulsive response is effectively released, and after passing over the hill, the system proceeds toward the region $r \gg R$, where $\delta(r) \rightarrow 0$ asymptotically.

The Wormhole Marginal Throat Selected by a Minimal Complex-Conjugate Scale and Its Opening Conditions

Misaki Kasai 

(Dated: June 10, 2026)

ABSTRACT

We investigate the opening conditions of the wormhole marginal throat selected by a minimal complex-conjugate scale. Starting from the dimensionless scale ratio $x = r/R$, the pair $\zeta_{\pm} = x \pm i$ generates a real geometric response through phase cancellation. This response has its maximum at the characteristic scale $r = R$. In a Morris–Thorne-type static and spherically symmetric geometry, the pure geometric sector gives $b'(R) = 1$, meaning that the response selects a marginal throat rather than an opened one. To open this throat, we introduce a throat-opening tensor constructed from the variational residual window of the scalar-field quasi-solution. The resulting local conditions lower the throat derivative below unity while keeping the redshift function finite at the throat. For the representative choice $\alpha = \delta_{\max}$, we obtain $b'(R) \simeq 0.615$, satisfying the flare-out condition.

We further show that the radial matching coefficient can be fixed by the supply peak of the normalized exchange current, giving $\beta \simeq 1.739$. With this value, the exterior matching condition admits a regular inner branch at $x_a \simeq 2.416$. The outer branch at $x_a \simeq 4.001$ is rejected because the radial metric factor vanishes earlier at $x_h \simeq 3.217$, whereas the accepted inner branch satisfies the thin-shell-free matching condition $\Phi'_{\text{in}}(x_a) = \tilde{b}(x_a)/(2x_a^2 D(x_a)) = \Phi'_{\text{out}}(x_a) \simeq 2.126$ in dimensionless notation. On this branch, the redshift function and curvature invariants remain finite, with the matching surface located at $a \simeq 2.416R$ ($R = GM/c^2$), between the exterior Schwarzschild radius $r_{s,\text{out}} \simeq 2.201R$ and the exterior photon-sphere radius $r_{\text{ph},\text{out}} \simeq 3.302R$, satisfying $r_{s,\text{out}} < a < r_{\text{ph},\text{out}}$. Thus, the locally opened throat can be matched to a Schwarzschild exterior branch while retaining the exterior photon-sphere structure. Within the static and spherically symmetric setting, this regular branch can be extended across the throat in a proper-distance coordinate, giving a symmetric two-sided opened-throat candidate with identical matching radii $a \simeq 2.416R$ and exterior Schwarzschild radii $r_{s,\text{out}} \simeq 2.201R$ on both sides. These results show that the variational residual can function as an internal geometric window for opening the marginal throat selected by the minimal complex-conjugate scale, and that the resulting opened-throat region admits a regular two-sided exterior matching candidate.

I. INTRODUCTION

Wormhole geometries have long served as useful theoretical laboratories for exploring finite-radius throat structures and possible departures from classical black-hole interiors [1–4]. In classical general relativity, gravitational collapse of a sufficiently compact object leads to the formation of a black hole. In the classical description, the interior evolution approaches a curvature singularity at $r = 0$, signaling the breakdown of the classical theory. It is therefore natural to ask whether this central singularity can be replaced by a finite-radius geometric structure. Related attempts include gravastars [5, 6], regular black holes [7–10], black-bounce geometries [11–13], and quantum-corrected black-hole interiors [14–16].

A wormhole throat also provides one possible finite-radius structure. In the Morris–Thorne construction, a static and spherically symmetric geometry is described by a shape function $b(r)$ and a redshift function $\Phi(r)$ [17–19]. If the throat is located at $r = r_0$, the basic throat condition is $b(r_0) = r_0$, while the flare-out condition is $b'(r_0) < 1$, and the redshift factor must also remain finite at the throat, $0 < e^{2\Phi(r_0)} < \infty$ [17–21]. These conditions distinguish a physically opened throat from a formal radial minimum by requiring both flare-out and a finite redshift function. More general discussions of wormhole throat geometry, flare-out conditions, and energy-condition constraints can be found in Refs. [20–24].

The purpose of the present paper is not to assume an opened wormhole throat from the beginning. Instead, we ask what type of throat structure is selected by a minimal complex-conjugate scale defined in a radial one-dimensional scale sector. Starting from the dimensionless scale ratio $x = r/R$, we introduce the pair $\zeta_{\pm} = x \pm i$. This pair is minimal in the sense that it is the smallest complex-conjugate structure needed to generate a real geometric response through phase cancellation, without introducing an additional dimensionless imaginary-scale parameter. In this construction, the radial one-dimensional scale sector $x = r/R$ is treated as the primary structure from which the geometric response is generated, while the static and spherically symmetric spacetime geometry is used as the local geometric arena in which the resulting throat conditions are evaluated.

The key observation is that the pure geometric response derived from this minimal complex-conjugate scale does not directly produce an opened throat. For the Morris–Thorne-type geometry, the temporal component of the pure sector gives $b'(r) = r^2 \Lambda(r)$. Since the response reaches $\Lambda(R) = 1/R^2$ at the characteristic scale $r = R$, one obtains $b'(R) = 1$. Thus, the pure response saturates the flare-out condition and selects a marginal throat rather than an opened one.

The central question of this paper is therefore how this marginal throat can be opened without introduc-

ing an external exotic matter source, unlike the usual exotic-matter support associated with traversable wormholes [17–19, 24]. To address this question, we introduce a throat-opening tensor $\mathcal{Q}^\mu{}_\nu$. This tensor is interpreted as an internal geometric correction constructed using the variational residual window of the scalar-field quasi-solution. This viewpoint is distinct from thin-shell constructions, where the supporting stress-energy is localized on a junction hypersurface [25–27]. It is also distinct from approaches in which quantum stress-energy, vacuum polarization, semiclassical backreaction, or Casimir-type effects are used as effective sources for wormhole support [28–38]. We derive the local opening conditions $\mathcal{Q}^t{}_t(R) < 0$ and $\mathcal{Q}^r{}_r(R) = 0$, and show that the variational residual can lower the throat derivative from $b'(R) = 1$ to $b'(R) \simeq 0.615$, satisfying the flare-out condition locally.

The paper is organized as follows. We first derive the geometric response from the minimal complex-conjugate scale and formulate the variational residual window. We then show that the pure geometric sector selects a marginal throat and derive the local conditions required to open it. Finally, we construct the throat-opening tensor from the residual window, analyze the exterior matching branch, and discuss the resulting static two-sided opened-throat candidate.

This organization also reflects the scope of the present work. We do not claim a fully dynamical traversable wormhole solution in this paper. Rather, we show that the minimal complex-conjugate scale selects a wormhole marginal throat, and that a variational-residual-based throat-opening tensor provides the local conditions under which this marginal throat can be opened. Within the static and spherically symmetric setting, the resulting opened-throat branch can be extended to a symmetric two-sided candidate and matched to a regular Schwarzschild exterior branch. Schwarzschild-like wormholes provide useful comparison cases for this type of exterior matching [1, 3, 4], while black-hole mimicker and dark-compact-object phenomenology provides the broader observational context [39]. A full dynamical stability analysis under time-dependent perturbations is left for future work, together with possible extensions to wave scattering and echo phenomenology [40–44], as well as gray-body-factor calculations and related numerical tools [45].

II. MINIMAL COMPLEX-CONJUGATE SCALE AND GEOMETRIC RESPONSE

We begin with the radial one-dimensional scale ratio

$$x = \frac{r}{R},$$

where R denotes the positive characteristic length scale of each physical regime, while both r and R are treated as dimensionless quantities until calibrated against obser-

vations. The dimensionless minimal complex-conjugate scale pair is defined by

$$\zeta_\pm = x \pm i.$$

This pair is symmetric with respect to the real x -axis and introduces the smallest imaginary displacement away from the real scale sector.

The real background response is obtained by adding the logarithms of the two conjugate branches,

$$\mathcal{B}(x) = \ln \zeta_+ + \ln \zeta_-.$$

Since $\zeta_+\zeta_- = 1 + x^2$, this gives

$$\mathcal{B}(x) = \ln(1 + x^2).$$

Thus, the imaginary phases cancel, and the remaining background response is real.

The scalar profile is defined as the radial derivative of this real background response,

$$\phi(r) = \frac{d}{dr} \ln \left(1 + \frac{r^2}{R^2} \right).$$

Therefore,

$$\phi(r) = \frac{2r}{R^2 + r^2}.$$

This profile is regular at $r = 0$, preserves the characteristic scale R , and decreases as $2/r$ for $r \gg R$.

The radius-dependent geometric response is defined by

$$\Lambda(r) = \phi(r)^2.$$

Thus,

$$\Lambda(r) = \frac{4r^2}{(R^2 + r^2)^2}.$$

Equivalently, using $x = r/R$, this can be written as

$$\Lambda(r) = \frac{1}{R^2} F(x), \quad F(x) = \frac{4x^2}{(1 + x^2)^2}.$$

The function $F(x)$ reaches its maximum at $x = 1$, where

$$F(1) = 1.$$

Therefore,

$$\Lambda(R) = \frac{1}{R^2}.$$

This peak relation $\Lambda(R) = 1/R^2$ is the key input for the throat analysis below.

In the black-hole interpretation, the same characteristic scale can be identified with the critical radius at which the inward gravitational acceleration is balanced

by the scale-dependent repulsive response. Let the repulsive acceleration associated with the geometric response be proportional to the scalar profile,

$$a_{\text{rep}}(r) = c_0 \phi(r) = \frac{2c_0 r}{R^2 + r^2}.$$

Here, c_0 is a characteristic scale coefficient that converts the scalar profile into an acceleration, with dimension m^2/s^2 , equivalent to an energy per unit mass. This acceleration reaches its maximum at $r = R$. We therefore identify the critical radius with the characteristic scale,

$$r_{\text{crit}} = R.$$

At the critical radius, the repulsive acceleration is

$$a_{\text{rep}}(r_{\text{crit}}) = \frac{c_0}{r_{\text{crit}}}.$$

Balancing this with the Newtonian gravitational acceleration GM/r^2 at the critical radius gives

$$\frac{c_0}{r_{\text{crit}}} = \frac{GM}{r_{\text{crit}}^2},$$

and therefore

$$c_0 = \frac{GM}{r_{\text{crit}}}.$$

If the repulsive energy scale at the collapse-stopping radius is identified with the black-hole mass-energy scale, one has $c_0 = c^2$. Hence,

$$r_{\text{crit}} = \frac{GM}{c^2}.$$

Since $r_{\text{crit}} = R$, in the black-hole interior interpretation,

$$R = \frac{GM}{c^2} = \frac{1}{2}r_s, \quad r_s = \frac{2GM}{c^2},$$

where r_s is the Schwarzschild radius [46].

This identification does not mean that $R = GM/c^2$ is already an opened throat. It only fixes the characteristic black-hole interior scale selected by the geometric response. As shown later, the pure geometric sector first gives a marginal throat at this scale, and the opening condition must be supplied separately.

III. VARIATIONAL RESIDUAL WINDOW

In this section, the scalar profile $\phi(r) = 2r/(R^2 + r^2)$ in the one-dimensional radial scale sector is substituted into the corresponding standard one-dimensional Lagrangian. The resulting deviation from the Euler-Lagrange extremum condition is read as the variational residual enclosing the internal response tension.

We use the standard kinetic-minus-potential form

$$\mathcal{L}_\phi = -\frac{1}{2} \left(\frac{d\phi}{dr} \right)^2 - V(\phi).$$

The potential is chosen as the minimal even polynomial containing the leading quadratic response and the first nonlinear correction,

$$V(\phi) = -\phi^2 + A\phi^4.$$

The coefficient A is fixed by requiring the quadratic and quartic contributions to belong to the same characteristic $4/R^2$ hierarchy after substituting the scale-preserving profile.

Introducing $x = r/R$, the quadratic contribution becomes

$$\phi(r)^2 = \frac{4r^2}{(R^2 + r^2)^2} = \frac{4}{R^2} \frac{x^2}{(1 + x^2)^2}.$$

Thus, the leading quadratic response belongs to the overall $4/R^2$ hierarchy.

The quartic contribution is

$$\phi(r)^4 = \left[\frac{4}{R^2} \frac{x^2}{(1 + x^2)^2} \right]^2 = \left(\frac{4}{R^2} \right)^2 \left[\frac{x^2}{(1 + x^2)^2} \right]^2.$$

This term contains one additional factor of $4/R^2$ compared with the quadratic response. Therefore, to place the quartic correction in the same $4/R^2$ hierarchy as the leading term, we choose

$$A = \frac{1}{4}R^2.$$

With this choice,

$$A\phi(r)^4 = \frac{1}{4}R^2 \left(\frac{4}{R^2} \right)^2 \left[\frac{x^2}{(1 + x^2)^2} \right]^2 = \frac{4}{R^2} \left[\frac{x^2}{(1 + x^2)^2} \right]^2.$$

Hence, the quartic correction is placed in the same overall $4/R^2$ hierarchy as the leading quadratic term.

The scalar potential is therefore fixed as

$$V(\phi) = -\phi^2 + \frac{1}{4}R^2\phi^4.$$

After substituting the scalar profile, this becomes

$$V(\phi(r)) = -\frac{4}{R^2} \frac{x^2}{(1 + x^2)^2} + \frac{4}{R^2} \left[\frac{x^2}{(1 + x^2)^2} \right]^2.$$

The first term is the leading response $-\Lambda(r)$, while the second term is the first nonlinear correction within the same $4/R^2$ scale hierarchy.

The corresponding Euler-Lagrange extremum condition (with r and R still being dimensionless quantities) is

$$\frac{d^2\phi}{dr^2} = R^2\phi^3 - 2\phi.$$

We define the variational residual by

$$\delta(r) \equiv \frac{d^2\phi}{dr^2} - (R^2\phi^3 - 2\phi).$$

Substituting the scalar profile into this definition gives

$$\delta(r) = \frac{4r(R^4 - 3R^2 + r^4 + r^2)}{(R^2 + r^2)^3}.$$

This residual measures the deviation of the scale-preserving scalar profile from the Euler–Lagrange extremum condition. In this sense, it can also be viewed as a reduced, dimensionless composition profile encoding the tension between the kinetic-type and potential-type response components. It is used to localize the internal opening correction.

To select a representative dimensionless residual profile, we require the residual to have a smoother onset at the origin by eliminating its linear term. The first derivative at the origin is

$$\left. \frac{d\delta}{dr} \right|_{r=0} = \frac{4(R^2 - 3)}{R^4}.$$

Thus, imposing

$$\left. \frac{d\delta}{dr} \right|_{r=0} = 0$$

gives

$$R^2 = 3.$$

Taking the positive branch gives $R = \sqrt{3}$, and we denote this selected reduced-coordinate value by $R_{(\sqrt{3})} \equiv \sqrt{3}$. With this value, the residual becomes

$$\delta(r) = \frac{4r^3(r^2 + 1)}{(r^2 + 3)^3}.$$

This form isolates a finite, localized residual structure while keeping the characteristic scale explicit.

Numerically, this residual has the values

$$\delta(R_{(\sqrt{3})}) \simeq 0.385, \quad \delta_{\max} \simeq 0.641.$$

The maximum occurs at approximately $r \simeq 3.56$. Therefore, the residual is finite at the throat scale and reaches its peak outside the throat.

Using the relation $x = r/R$, the normalized residual window is defined by

$$W_\delta(x) = \left. \frac{\delta(Rx)}{\delta_{\max}} \right|_{R=R_{(\sqrt{3})}}.$$

At the throat scale $x = 1$, this gives

$$W_\delta(1) = \frac{\delta(R_{(\sqrt{3})})}{\delta_{\max}} \simeq \frac{0.385}{0.641} \simeq 0.601.$$

The role of $W_\delta(x)$ is to specify where the opening correction is localized. Since $W_\delta(x)$ is dimensionless, the product $W_\delta(x)\Lambda(r)$ has the same curvature dimension as the Einstein tensor. Thus, the variational residual window can be used to construct a throat-opening tensor without treating the residual itself as an independently introduced exotic matter source. The residual window constructed here will be used below as the localization structure of the throat-opening tensor.

IV. MARGINAL THROAT SELECTED BY THE PURE GEOMETRIC SECTOR

We now examine the throat structure selected by the pure geometric response. In this section, no throat-opening tensor is introduced. The purpose is to determine what kind of throat is obtained from $\Lambda(r)$ alone.

We use the Morris–Thorne-type static and spherically symmetric line element [17, 19]

$$ds^2 = -e^{2\Phi(r)}c^2dt^2 + \frac{dr^2}{1 - b(r)/r} + r^2d\Omega^2.$$

Here, $b(r)$ is the shape function and $\Phi(r)$ is the redshift function. If the throat is located at $r = r_0$, the throat condition is

$$b(r_0) = r_0.$$

In the present construction, the characteristic scale selected by the geometric response is identified with the throat candidate,

$$r_0 = R.$$

Thus, the throat condition becomes

$$b(R) = R.$$

The pure geometric sector is written as

$$G^\mu{}_\nu + \Lambda(r)\delta^\mu{}_\nu = 0.$$

For the Morris–Thorne metric, the temporal mixed component is

$$G^t{}_t = -\frac{b'(r)}{r^2}.$$

Therefore, the temporal component of the pure sector gives

$$-\frac{b'(r)}{r^2} + \Lambda(r) = 0.$$

Equivalently,

$$b'(r) = r^2\Lambda(r).$$

Evaluating this relation at the candidate throat $r = R$, we obtain

$$b'(R) = R^2\Lambda(R).$$

Since the response generated by the minimal complex-conjugate scale satisfies $\Lambda(R) = 1/R^2$, this gives

$$b'(R) = 1.$$

The Morris–Thorne flare-out condition for an opened throat [17, 20, 21] is

$$b'(R) < 1.$$

The pure geometric response therefore does not satisfy the strict flare-out inequality. Instead, it reaches the boundary value $b'(R) = 1$, saturating the flare-out condition as an equality.

Thus, the pure geometric response selects a marginal throat at $r = R$. It determines the throat scale but does not open the throat by itself.

V. STATIC ACTION NOTATION FOR THE OPENING TENSOR

Before specifying the local throat-opening components, we formulate the opening tensor in an action notation adapted to the static and spherically symmetric geometry. In this notation, the opening tensor is generated by a scalar quantity whose metric variation produces the anisotropic temporal, radial, and transverse responses required below.

We first express the same dimensionless scale variable x geometrically by using the area $\mathcal{A}$ of the symmetry two-spheres,

$$x = \frac{1}{R} \sqrt{\frac{\mathcal{A}}{4\pi}}.$$

In the areal-radius coordinate used in the following sections, $\mathcal{A} = 4\pi r^2$, and therefore this definition reduces to $x = r/R$.

Since the radial sector enters $\mathcal{Q}_{\mu\nu}$ only through $s_\mu s_\nu$, we use the local representation of the radial unit covector

$$s_\mu = \frac{\nabla_\mu x}{\sqrt{g^{\lambda\eta} \nabla_\lambda x \nabla_\eta x}}.$$

With this representation, the normalization condition is

$$g^{\mu\nu} s_\mu s_\nu = 1.$$

In the static sector, the temporal unit vector is defined from the static Killing vector ξ^μ as

$$u^\mu = \frac{\xi^\mu}{\sqrt{-\xi^\lambda \xi_\lambda}}.$$

It satisfies

$$u^\mu u_\mu = -1.$$

The static and spherically symmetric background also gives the orthogonality condition

$$u^\mu s_\mu = 0.$$

The vectors u^μ and s^μ define the local temporal and radial directions of the static spherical geometry. The mixed components of $\mathcal{Q}^\mu{}_\nu$ are treated as scalar functions of x in this sector. A diagonal anisotropic opening tensor compatible with spherical symmetry can therefore be written in the projector form

$$\mathcal{Q}_{\mu\nu} = \mathcal{Q}^\theta{}_\theta g_{\mu\nu} - (\mathcal{Q}^t{}_t - \mathcal{Q}^\theta{}_\theta) u_\mu u_\nu - (\mathcal{Q}^\theta{}_\theta - \mathcal{Q}^r{}_r) s_\mu s_\nu.$$

In the mixed-index form adapted to the static spherical background, this expression gives

$$\mathcal{Q}^\mu{}_\nu = \text{diag}(\mathcal{Q}^t{}_t, \mathcal{Q}^r{}_r, \mathcal{Q}^\theta{}_\theta, \mathcal{Q}^\theta{}_\theta).$$

Thus the temporal, radial, and transverse opening responses are organized by the natural projectors of the static spherical sector.

We now introduce an action notation that generates this projector form. Let

$$S_Q = \frac{1}{\kappa} \int d^4x \sqrt{-g} \Xi,$$

where Ξ is the scalar generating quantity for the opening tensor in the static spherical sector. We take

$$\Xi = \mathcal{Q}^\theta{}_\theta + \frac{1}{2} (\mathcal{Q}^t{}_t - \mathcal{Q}^\theta{}_\theta) (I_u + 1) + \frac{1}{2} (\mathcal{Q}^\theta{}_\theta - \mathcal{Q}^r{}_r) (I_s - 1),$$

with

$$I_u = g^{\mu\nu} u_\mu u_\nu, \quad I_s = g^{\mu\nu} s_\mu s_\nu.$$

On the static spherical background, these invariants reduce to

$$I_u = -1, \quad I_s = 1.$$

Therefore the background value of the generating scalar is

$$\Xi_0 = \mathcal{Q}^\theta{}_\theta.$$

For this static-sector action notation, the algebraic metric variation is taken with respect to the projector invariants I_u and I_s . The opening tensor is defined by the metric variation of S_Q ,

$$\mathcal{Q}_{\mu\nu} = -\frac{2\kappa}{\sqrt{-g}} \frac{\delta S_Q}{\delta g^{\mu\nu}}.$$

Since Ξ depends algebraically on the metric through I_u and I_s in the present sector, this becomes

$$\mathcal{Q}_{\mu\nu} = g_{\mu\nu} \Xi - 2 \frac{\partial \Xi}{\partial g^{\mu\nu}}.$$

Using

$$\frac{\partial I_u}{\partial g^{\mu\nu}} = u_\mu u_\nu,$$

and

$$\frac{\partial I_s}{\partial g^{\mu\nu}} = s_\mu s_\nu,$$

one obtains, on the static spherical background,

$$\mathcal{Q}_{\mu\nu} = \mathcal{Q}^\theta{}_\theta g_{\mu\nu} - (\mathcal{Q}^t{}_t - \mathcal{Q}^\theta{}_\theta) u_\mu u_\nu - (\mathcal{Q}^\theta{}_\theta - \mathcal{Q}^r{}_r) s_\mu s_\nu.$$

Thus the anisotropic opening tensor used below is represented by a scalar action notation in the static and spherically symmetric sector.

At this stage, the three scalar components $\mathcal{Q}^t{}_t(x)$, $\mathcal{Q}^r{}_r(x)$, and $\mathcal{Q}^\theta{}_\theta(x)$ have not yet been specified. The next section derives the local throat conditions on the temporal and radial components. The residual-window construction then determines the minimal forms of $\mathcal{Q}^t{}_t(x)$ and $\mathcal{Q}^r{}_r(x)$, while the transverse component $\mathcal{Q}^\theta{}_\theta(x)$ is fixed by the covariant conservation of the combined effective geometric sector.

A full time-dependent perturbation theory would require specifying how the temporal and radial projectors are extended away from the static sector. That issue is left for future work together with the dynamical stability analysis.

VI. LOCAL OPENING CONDITIONS AT THE MARGINAL THROAT

Having formulated the opening tensor in an action notation adapted to the static and spherically symmetric sector, we now derive the local conditions required for this tensor to open the marginal throat selected by the pure geometric response.

As shown above, the pure sector gives $b'(R) = 1$ at the marginal throat located at $r = R$. Hence the Morris–Thorne flare-out condition [17, 20, 21] is saturated rather than strictly satisfied. The role of the opening tensor $\mathcal{Q}^\mu{}_\nu$ is to deform this marginal configuration into a locally opened throat.

We write the corrected geometric field equation as

$$G^\mu{}_\nu + \Lambda(r)\delta^\mu{}_\nu + \mathcal{Q}^\mu{}_\nu(r) = 0,$$

leaving the detailed functional form of the throat-opening tensor $\mathcal{Q}^\mu{}_\nu$ unspecified at this stage. We only derive the local conditions that this tensor must satisfy at the throat.

For the Morris–Thorne-type metric, the temporal mixed component is

$$G^t_t = -\frac{b'(r)}{r^2}.$$

Therefore, the temporal component of the corrected equation gives

$$-\frac{b'(r)}{r^2} + \Lambda(r) + \mathcal{Q}^t_t(r) = 0.$$

Equivalently,

$$b'(r) = r^2 [\Lambda(r) + \mathcal{Q}^t_t(r)].$$

At the candidate throat $r = R$, using $\Lambda(R) = 1/R^2$, this becomes

$$b'(R) = 1 + R^2 \mathcal{Q}^t_t(R).$$

To open the marginal throat, the flare-out condition requires $b'(R) < 1$. Hence,

$$1 + R^2 \mathcal{Q}^t_t(R) < 1,$$

and therefore

$$\mathcal{Q}^t_t(R) < 0.$$

This is the first local opening condition. It states that the temporal component of the opening tensor must give a negative correction at the throat in order to lower $b'(R)$ below unity.

Next, we consider the redshift function. The radial component of the corrected equation gives

$$-\frac{b(r)}{r^3} + 2 \left(1 - \frac{b(r)}{r}\right) \frac{\Phi'(r)}{r} + \Lambda(r) + \mathcal{Q}^r_r(r) = 0.$$

Solving this relation for $\Phi'(r)$, we obtain

$$\Phi'(r) = \frac{r}{2(1 - b(r)/r)} \left[\frac{b(r)}{r^3} - \Lambda(r) - \mathcal{Q}^r_r(r) \right].$$

At the throat, $b(R) = R$, so the denominator $1 - b(R)/R$ vanishes. For the redshift function to remain finite at the throat, the numerator must vanish at the same point [17, 19]. Evaluating the numerator at $r = R$, we find

$$\frac{b(R)}{R^3} - \Lambda(R) - \mathcal{Q}^r_r(R) = \frac{1}{R^2} - \frac{1}{R^2} - \mathcal{Q}^r_r(R).$$

Thus, the regularity condition reduces to

$$\mathcal{Q}^r_r(R) = 0.$$

This is the second local opening condition. It ensures that the redshift function does not acquire a divergence at the throat.

The two local conditions required to open the marginal throat are therefore

$$\mathcal{Q}^t_t(R) < 0, \quad \mathcal{Q}^r_r(R) = 0.$$

The first condition controls the flare-out inequality, while the second controls the finiteness of the redshift function. The following section constructs a concrete form of $\mathcal{Q}^\mu{}_\nu$ from the variational residual window and shows that these two conditions can be satisfied simultaneously.

VII. OPENING BY THE VARIATIONAL RESIDUAL WINDOW

We now construct the throat-opening tensor using the variational residual window introduced above. The purpose is to deform the marginal throat selected by the pure geometric sector into a locally opened throat.

The required local opening conditions are $\mathcal{Q}^t_t(R) < 0$ and $\mathcal{Q}^r_r(R) = 0$. A minimal form satisfying the first condition is obtained by choosing the temporal component as

$$\mathcal{Q}^t_t(r) = -\alpha W_\delta(x) \Lambda(r),$$

where α is a dimensionless coefficient controlling the strength of the temporal opening component, with $0 \leq \alpha \leq \delta_{\max}$. At the throat $x = 1$, this gives

$$\mathcal{Q}^t_t(R) = -\alpha W_\delta(1) \Lambda(R) = -\frac{\alpha W_\delta(1)}{R^2}.$$

Therefore, for $\alpha > 0$ and $W_\delta(1) > 0$, the condition $\mathcal{Q}^t_t(R) < 0$ is satisfied.

The corrected temporal equation gives

$$b'(r) = r^2 [\Lambda(r) + \mathcal{Q}^t_t(r)].$$

Evaluating this relation at $r = R$, and using $\Lambda(R) = 1/R^2$, one obtains

$$b'(R) = 1 - \alpha W_\delta(1).$$

Thus, the marginal value $b'(R) = 1$ is lowered below unity whenever $\alpha W_\delta(1) > 0$. If one additionally requires $0 < b'(R) < 1$, then the coefficient must satisfy $0 < \alpha W_\delta(1) < 1$.

For the radial component, the redshift regularity condition requires $\mathcal{Q}^r_r(R) = 0$. We use the form

$$\mathcal{Q}^r_r(r) = -\beta \left(1 - \frac{1}{x}\right) W_\delta(x) \Lambda(r),$$

where β is a dimensionless coefficient controlling the strength of the radial opening component. Since $1 - 1/x = 0$ at $x = 1$, this component automatically satisfies

$$\mathcal{Q}^r_r(R) = 0.$$

Therefore, the temporal component opens the throat, while the radial component is constructed so as to preserve the finiteness of the redshift function at the throat.

Equivalently, using $\Lambda(r) = R^{-2}F(x)$, these components can be written in dimensionless form as

$$R^2 \mathcal{Q}^t_t(x) = -\alpha W_\delta(x) F(x),$$

and

$$R^2 \mathcal{Q}^r_r(x) = -\beta \left(1 - \frac{1}{x}\right) W_\delta(x) F(x).$$

We now evaluate the opening strength. Since $W_\delta(1) = \delta(R_{(\sqrt{3})})/\delta_{\max}$, a natural choice is to identify the opening coefficient with the maximum residual strength,

$$\alpha = \delta_{\max}.$$

Then

$$\alpha W_\delta(1) = \delta_{\max} \frac{\delta(R_{(\sqrt{3})})}{\delta_{\max}} = \delta(R_{(\sqrt{3})}),$$

and therefore

$$b'(R) = 1 - \delta(R_{(\sqrt{3})}).$$

Using $\delta(R_{(\sqrt{3})}) \simeq 0.385$, we obtain

$$b'(R) \simeq 0.615.$$

Hence,

$$0 < b'(R) < 1$$

is satisfied.

This result shows that the variational residual window can deform the marginal throat into a locally opened throat. This residual window is not interpreted as an independently introduced exotic matter source. Rather, it acts as a dimensionless localization window multiplying the geometric response $\Lambda(r)$. In this sense, the quasi-solution character of the scalar profile supplies the internal geometric correction required to open the marginal throat.

The functional form of the transverse component $\mathcal{Q}^\theta_\theta$ is also not specified independently, but is fixed by covariant conservation, as discussed later.

VIII. EXCHANGE CURRENT AND RADIAL MATCHING COEFFICIENT

The radial component of the throat-opening tensor was introduced with a dimensionless coefficient β . In this section, we fix this coefficient from the exchange-current structure of the scale-dependent geometric response, rather than treating it as an independent free parameter.

The geometric response can be written in the universal form

$$\Lambda(r) = \frac{1}{R^2} F(x), \quad F(x) = \frac{4x^2}{(1+x^2)^2}, \quad x = \frac{r}{R}.$$

Since $\Lambda(r)$ depends on the radial coordinate, its radial gradient defines an exchange current associated with the scale-dependent geometric sector. In the static and spherically symmetric case, we write

$$J_r = \frac{1}{\kappa} \frac{d\Lambda}{dr},$$

where $\kappa = 8\pi G/c^4$. Using $x = r/R$, one obtains

$$\frac{d\Lambda}{dr} = \frac{1}{R^3} \frac{dF}{dx},$$

and therefore

$$J_r = \frac{1}{\kappa R^3} \frac{dF}{dx}.$$

The dimensionless exchange current is then obtained by normalizing J_r ,

$$\tilde{J}(x) \equiv \kappa R^3 J_r = \frac{dF}{dx},$$

and explicitly

$$\tilde{J}(x) = \frac{8x(1-x^2)}{(1+x^2)^3}.$$

This function changes sign at $x = 1$. For $x < 1$, $\tilde{J}(x) > 0$, while for $x > 1$, $\tilde{J}(x) < 0$. Thus, $x = 1$ is the transition point between the supply and recovery branches of the exchange-current structure.

The extrema of the normalized exchange current are determined by

$$\frac{d\tilde{J}}{dx} = \frac{d^2F}{dx^2} = 0.$$

The second derivative is

$$\frac{d^2F}{dx^2} = \frac{8(3x^4 - 8x^2 + 1)}{(1+x^2)^4}.$$

Therefore, the extrema satisfy

$$3x^4 - 8x^2 + 1 = 0.$$

Solving this equation gives

$$x_{\text{sup}} = \sqrt{\frac{4 - \sqrt{13}}{3}}, \quad x_{\text{rec}} = \sqrt{\frac{4 + \sqrt{13}}{3}}.$$

Numerically,

$$x_{\text{sup}} \simeq 0.363, \quad x_{\text{rec}} \simeq 1.592.$$

Substitution into $\tilde{J}(x)$ yields

$$\tilde{J}(x_{\text{sup}}) \simeq +1.739, \quad \tilde{J}(x_{\text{rec}}) \simeq -0.443.$$

The positive extremum at x_{sup} represents the maximum supply branch of the exchange current. We use this supply peak to fix the radial matching coefficient,

$$\beta = \tilde{J}_{\text{sup}}^{\text{max}} \simeq 1.739.$$

With this prescription, β is not introduced as an arbitrary parameter. It is determined by the maximum exchange capacity of the scale-dependent response itself.

The recovery peak $\tilde{J}(x_{\text{rec}}) \simeq -0.443$ characterizes the outer recovery branch, where the exchange current changes sign and the effective correction is returned to the geometric sector. In the present throat-opening construction, the supply peak fixes the radial matching strength, while the recovery branch provides the scale at which the correction begins to relax toward the exterior region.

Finally, we note that the exchange current is balanced over the full radial domain. Since

$$\Lambda(0) = \Lambda(\infty) = 0,$$

one has

$$\int_0^\infty J_r dr = \frac{1}{\kappa} [\Lambda(\infty) - \Lambda(0)] = 0.$$

Thus, the one-dimensional radial exchange is balanced between the supply and recovery branches.

In addition to this radial balance, the corresponding spherical flux is

$$\Phi_J(r) = \oint_{S_r} J^i d\Sigma_i = 4\pi r^2 J_r.$$

Using

$$J_r = \frac{1}{\kappa} \frac{d}{dr} \left[\frac{4r^2}{(R^2 + r^2)^2} \right] = \frac{1}{\kappa} \frac{8r(R^2 - r^2)}{(R^2 + r^2)^3},$$

we obtain

$$\Phi_J(r) = \frac{32\pi}{\kappa} \frac{r^3(R^2 - r^2)}{(R^2 + r^2)^3}.$$

The endpoint limits are

$$\Phi_J(0) = 0, \quad \Phi_J(\infty) = 0.$$

Therefore, since $\Phi_J(0) = \Phi_J(\infty) = 0$, the spherical flux has no net endpoint contribution. In this sense, the exchange current represents an internal redistribution between the scale-dependent geometric response and the opening correction through the supply and recovery branches.

At sufficiently large characteristic scales, the exchange current is strongly suppressed. For a cosmological scale $R \sim 10^{26} \text{ m}$, the estimate $|d\Lambda/dr| \sim R^{-3} \sim 10^{-78} \text{ m}^{-3}$ shows that $\Lambda(r)$ can be treated as approximately constant. Hence, in the large-scale limit, the local exchange current becomes negligible and the usual Friedmann dynamics with a constant Λ is effectively recovered [47, 48].

IX. COVARIANT CONSERVATION AND TRANSVERSE RESPONSE

In the present construction, the throat-opening tensor is interpreted as an effective correction to the geometric response, rather than as an independently assumed matter source. Since the scale-dependent response $\Lambda(r)$ depends on the radial coordinate, the term $\Lambda(r)\delta^\mu_\nu$ is not separately conserved in general. Therefore, we impose covariant conservation on the combined effective geometric sector,

$$\nabla_\mu [\Lambda(r)\delta^\mu_\nu + \mathcal{Q}^\mu_\nu] = 0.$$

Equivalently, the opening tensor satisfies

$$\nabla_\mu \mathcal{Q}^\mu_\nu = -\partial_\nu \Lambda(r).$$

Thus, $\mathcal{Q}^\mu_\nu$ can be interpreted as an effective geometric correction that compensates for the radial variation of $\Lambda(r)$, rather than as an independently conserved matter tensor.

For a static and spherically symmetric correction sector, we take the opening tensor to be diagonal,

$$\mathcal{Q}^\mu_\nu = \text{diag}(\mathcal{Q}^t_t, \mathcal{Q}^r_r, \mathcal{Q}^\theta_\theta, \mathcal{Q}^\theta_\theta).$$

The two angular components are equal by spherical symmetry, so there is only one independent transverse component.

The nontrivial conservation equation is the radial component. It gives

$$\frac{d}{dr} (\Lambda + \mathcal{Q}^r_r) + \Phi'(r) (\mathcal{Q}^r_r - \mathcal{Q}^t_t) + \frac{2}{r} (\mathcal{Q}^r_r - \mathcal{Q}^\theta_\theta) = 0.$$

Solving this relation for the transverse component yields

$$\mathcal{Q}^\theta_\theta = \mathcal{Q}^r_r + \frac{r}{2} \left[\frac{d}{dr} (\Lambda + \mathcal{Q}^r_r) + \Phi'(r) (\mathcal{Q}^r_r - \mathcal{Q}^t_t) \right].$$

This expression shows that the angular component is not freely prescribed once $\mathcal{Q}^t_t$, $\mathcal{Q}^r_r$, and $\Phi(r)$ are specified. It is fixed by the covariant conservation of the combined effective sector.

At the throat, the radial opening component satisfies $\mathcal{Q}^r_r(R) = 0$. Also, since $\Lambda(r)$ reaches its maximum at $r = R$, one has $d\Lambda/dr|_{r=R} = 0$. Using

$$\mathcal{Q}^t_t(R) = -\frac{\alpha W_\delta(1)}{R^2}, \quad \left. \frac{d\mathcal{Q}^r_r}{dr} \right|_{r=R} = -\frac{\beta W_\delta(1)}{R^3},$$

the transverse component at the throat becomes

$$\mathcal{Q}^\theta_\theta(R) = \frac{W_\delta(1)}{2R^2} [-\beta + \alpha R\Phi'(R)].$$

Equivalently, since

$$R\Phi'(R) = \left. \frac{d\Phi}{dx} \right|_{x=1},$$

this throat value can be written in the dimensionless form

$$R^2 \mathcal{Q}^\theta_\theta(R) = \frac{W_\delta(1)}{2} \left[-\beta + \alpha \left. \frac{d\Phi}{dx} \right|_{x=1} \right].$$

This formula gives a local diagnostic of the transverse opening response. Since $W_\delta(1)$, α , and β are finite, a finite throat limit of $d\Phi/dx$ implies that $\mathcal{Q}^\theta_\theta(R)$ is also finite. Hence, the transverse component is not independently prescribed, but is fixed by covariant conservation once the temporal and radial opening components are specified, and this conservation condition does not require a singular or divergent transverse response at the opened throat. It is important to note that the transverse component need not vanish or have a positive sign. The only requirement is the finite dimensionless condition

$$|R^2 \mathcal{Q}^\theta_\theta(R)| < \infty.$$

X. EXTERIOR MATCHING AND REGULARITY

This section analyzes the matching of the locally opened throat region to an exterior Schwarzschild geometry. Using the black-hole interior identification $R = GM/c^2$ obtained above, let the matching surface be located at

$$a = Rx_a,$$

and define the dimensionless shape function by

$$\tilde{b}(x) = \frac{b(r)}{R}.$$

The throat condition is $\tilde{b}(1) = 1$. In the interior region, the temporal component of the corrected geometric equation gives

$$\tilde{b}'(x) = x^2 F(x) [1 - \alpha W_\delta(x)],$$

and therefore

$$\tilde{b}(x) = 1 + \int_1^x u^2 F(u) [1 - \alpha W_\delta(u)] du.$$

We take the exterior region to be described by the Schwarzschild line element [46],

$$ds_{\text{out}}^2 = -\left(1 - \frac{2GM_{\text{out}}}{c^2 r}\right) c^2 dt^2 + \frac{dr^2}{1 - \frac{2GM_{\text{out}}}{c^2 r}} + r^2 d\Omega^2.$$

Continuity of the radial metric component at $r = a$ requires

$$b_{\text{in}}(a) = \frac{2GM_{\text{out}}}{c^2}.$$

Thus the exterior mass is determined by

$$M_{\text{out}} = \frac{c^2}{2G} b_{\text{in}}(a) = \frac{c^2 R}{2G} \tilde{b}(x_a).$$

If the internal reference mass is defined by $R = GM/c^2$, then

$$\frac{M_{\text{out}}}{M} = \frac{\tilde{b}(x_a)}{2}.$$

The time component of the first fundamental form gives

$$e^{2\Phi_{\text{in}}(a)} = 1 - \frac{2GM_{\text{out}}}{c^2 a} = 1 - \frac{\tilde{b}(x_a)}{x_a}.$$

Therefore, the matching surface must satisfy

$$D(x_a) \equiv 1 - \frac{\tilde{b}(x_a)}{x_a} > 0.$$

In contrast to thin-shell junction constructions, where the surface stress-energy is determined by junction conditions [25–27], the non-Schwarzschild radial effective contribution must vanish at the matching surface:

$$\Lambda(a) + \mathcal{Q}^r_r(a) = 0.$$

Using

$$\mathcal{Q}^r_r(r) = -\beta \left(1 - \frac{1}{x}\right) W_\delta(x) \Lambda(r),$$

this becomes

$$\Lambda(a) \left[1 - \beta \left(1 - \frac{1}{x_a}\right) W_\delta(x_a) \right] = 0.$$

Since $\Lambda(a) \neq 0$ at finite a , this condition reduces to

$$\beta \left(1 - \frac{1}{x_a}\right) W_\delta(x_a) = 1.$$

With $\beta = \tilde{J}_{\text{sup}}^{\text{max}} \simeq 1.739$, this equation admits two numerical solutions,

$$x_a \simeq 2.416, \quad x_a \simeq 4.001.$$

However, not both branches are physically suitable. The radial metric factor is controlled by

$$D(x) = 1 - \frac{\tilde{b}(x)}{x}.$$

Numerically, $D(x)$ vanishes at

$$x_h \simeq 3.217.$$

Therefore, the outer branch $x_a \simeq 4.001$ lies beyond this zero and is not suitable as a regular matching surface for the opened-throat interior.

The inner branch satisfies

$$x_a \simeq 2.416 < x_h \simeq 3.217.$$

Hence $D(x) > 0$ is maintained throughout $1 < x \leq x_a$. At the matching surface on this branch, one finds

$$D(x_a) \simeq 0.0887 > 0.$$

Thus, the interior opened-throat region can be matched to a Schwarzschild exterior at

$$a \simeq 2.416R$$

without crossing an internal zero of the radial metric factor.

The regularity of this inner branch can be further checked by evaluating the redshift function and the main curvature invariants. From the radial field equation, the redshift derivative is

$$\Phi'(r) = \frac{r}{2[1 - b(r)/r]} \left[\frac{b(r)}{r^3} - \Lambda(r) - \mathcal{Q}^r_r(r) \right].$$

Equivalently, in terms of $x = r/R$, this equation becomes

$$\frac{d\Phi}{dx} = \frac{x}{2D(x)} \left[\frac{\tilde{b}(x)}{x^3} - F(x) - R^2 \mathcal{Q}^r_r(x) \right],$$

with $D(x) = 1 - \tilde{b}(x)/x$. At the throat, both the numerator and denominator vanish consistently because $\tilde{b}(1) = 1$, $F(1) = 1$, and $R^2 \mathcal{Q}^r_r(1) = 0$. In particular, near the throat, both vanish linearly in $x-1$, while the denominator has leading coefficient $2D'(1)$, with $D'(1) > 0$. Therefore, the redshift derivative has a finite throat limit. Numerically, for the regular inner branch, one finds

$$\left. \frac{d\Phi}{dx} \right|_{x=1+} \simeq -1.741.$$

The redshift function also remains finite from the throat to the matching surface. The integrated change is

$$\Phi(x_a) - \Phi(1) \simeq 0.635.$$

The additive constant of Φ is fixed by the time-component matching condition,

$$e^{2\Phi(x_a)} = D(x_a),$$

or equivalently,

$$\Phi(x_a) = \frac{1}{2} \ln D(x_a).$$

The interior profile is then obtained by integrating inward,

$$\Phi(x) = \Phi(x_a) - \int_x^{x_a} \frac{d\Phi}{du} du.$$

In terms of the dimensionless coordinate x , the interior metric is equivalently written as

$$ds^2 = -e^{2\Phi(x)} c^2 dt^2 + \frac{R^2 dx^2}{D(x)} + R^2 x^2 d\Omega^2.$$

Since this metric is smooth for $1 < x < x_a$, it is then sufficient to evaluate the curvature invariants at the end-point limits. Denoting the Ricci scalar by $\mathcal{R}$, one obtains

$$\mathcal{R}(1^+) \simeq \frac{1.90}{R^2}, \quad \mathcal{R}(x_a) \simeq -\frac{1.00}{R^2}.$$

The Ricci tensor squared remains finite, with

$$R_{\mu\nu} R^{\mu\nu}(1^+) \simeq \frac{1.42}{R^4}, \quad R_{\mu\nu} R^{\mu\nu}(x_a) \simeq \frac{0.743}{R^4}.$$

The Kretschmann scalar also remains finite:

$$R_{\mu\nu\rho\sigma} R^{\mu\nu\rho\sigma}(1^+) \simeq \frac{4.75}{R^4}, \quad R_{\mu\nu\rho\sigma} R^{\mu\nu\rho\sigma}(x_a) \simeq \frac{1.28}{R^4}.$$

Thus, the inner matching branch does not develop a curvature singularity in the interval $1 \leq x \leq x_a$. Together with $D(x) > 0$ for $1 < x \leq x_a$, these finite curvature invariants support the interpretation of this branch as a locally regular opened-throat region matched to a Schwarzschild exterior. The matching surface lies outside the exterior Schwarzschild radius determined through the internal shape function, while the detailed behavior of the photon sphere and the associated shadow scale can be examined separately.

XI. TWO-SIDED EXTENSION AND GLOBAL MATCHING CANDIDATE

We now examine whether the accepted inner branch can be extended across the throat to form a two-sided static geometry matched to Schwarzschild exterior regions on both sides.

For the representative inner branch, the numerical values relevant to the regular two-sided extension are

$$\tilde{b}'(1) \simeq 0.615 < 1, \quad D'(1) \simeq 0.385 > 0,$$

and, at the matching surface,

$$\tilde{b}(x_a) \simeq 2.201, \quad D(x_a) \simeq 0.0887 > 0.$$

Thus, the accepted branch satisfies $D(1) = 0$ at the throat and $D(x) > 0$ for $1 < x \leq x_a$. Hence, $D(x)$

has no additional horizon-like zero between the throat and the matching surface.

As discussed above, the non-Schwarzschild radial effective contribution must vanish at the matching surface:

$$\Lambda(a) + \mathcal{Q}^r_r(a) = 0.$$

Using $\Lambda(a) = R^{-2}F(x_a)$, this condition can be written in dimensionless form as

$$F(x_a) + R^2\mathcal{Q}^r_r(a) = 0.$$

On the other hand, the interior radial equation gives

$$\Phi'_{\text{in}}(x_a) = \frac{x_a}{2D(x_a)} \left[\frac{\tilde{b}(x_a)}{x_a^3} - F(x_a) - R^2\mathcal{Q}^r_r(a) \right].$$

Substituting the matching-surface condition into the interior expression, the last two terms cancel, and one obtains

$$\Phi'_{\text{in}}(x_a) = \frac{\tilde{b}(x_a)}{2x_a^2 D(x_a)}.$$

For the exterior Schwarzschild region, the first fundamental form gives

$$r_{s,\text{out}} = R\tilde{b}(x_a),$$

and therefore

$$\Phi'_{\text{out}}(x_a) = \frac{\tilde{b}(x_a)}{2x_a^2 D(x_a)}.$$

Thus,

$$\Phi'_{\text{in}}(x_a) = \frac{\tilde{b}(x_a)}{2x_a^2 D(x_a)} = \Phi'_{\text{out}}(x_a).$$

For the accepted matching radius $x_a \simeq 2.416$, this yields

$$\Phi'_{\text{in}}(x_a) = \Phi'_{\text{out}}(x_a) \simeq 2.126.$$

Therefore, the accepted inner branch satisfies the thin-shell-free matching condition, with the interior and exterior redshift gradients agreeing at the matching surface with the finite value $\simeq 2.126$.

A convenient coordinate for describing the two-sided extension is the proper radial distance l , defined by

$$\frac{dl}{dr} = \pm \left(1 - \frac{b(r)}{r} \right)^{-1/2}.$$

Introducing the dimensionless variables

$$x = \frac{r}{R}, \quad \tilde{b}(x) = \frac{b(r)}{R}, \quad D(x) = 1 - \frac{\tilde{b}(x)}{x},$$

the proper-distance relation becomes

$$\frac{dl}{R} = \pm \frac{dx}{\sqrt{D(x)}}.$$

The two signs correspond to the two sides of the throat. We choose the origin of l at the throat, so that

$$x = 1, \quad l = 0.$$

At this point,

$$D(1) = 0.$$

The flare-out condition obtained above gives

$$\tilde{b}'(1) < 1.$$

Since $D(x) = 1 - \tilde{b}(x)/x$, one finds at the throat

$$D'(1) = 1 - \tilde{b}'(1) > 0.$$

Therefore, near $x = 1$,

$$D(x) \simeq D'(1)(x - 1).$$

Substituting this into the proper-distance relation gives

$$l \simeq \pm 2R \sqrt{\frac{x - 1}{D'(1)}}.$$

Equivalently,

$$x - 1 \simeq \frac{D'(1)}{4} \left(\frac{l}{R} \right)^2.$$

Thus, the areal radius

$$r(l) = Rx(l)$$

is an even function of l near the throat and has a regular minimum at

$$l = 0.$$

In particular,

$$\left. \frac{dr}{dl} \right|_{l=0} = 0, \quad \left. \frac{d^2r}{dl^2} \right|_{l=0} > 0.$$

This gives the local proper-distance representation of the opened throat.

Since the inner branch satisfies

$$D(x) > 0 \quad (1 < x \leq x_a),$$

the proper distance from the throat to the matching surface is finite:

$$l_a = R \int_1^{x_a} \frac{dx}{\sqrt{D(x)}}.$$

Numerically, for the inner matching branch, this gives a finite value of order

$$l_a \simeq 5.5R.$$

Therefore, in the proper-distance coordinate, the interior region is represented as

$$-l_a \leq l \leq l_a,$$

with the throat at $l = 0$.

Following the standard picture of a symmetric two-sided wormhole throat [17, 19], a simple symmetric extension is obtained by taking

$$r(l) = r(-l), \quad \Phi(l) = \Phi(-l).$$

The interior metric can then be written in the proper-distance form

$$ds^2 = -e^{2\Phi(l)} c^2 dt^2 + dl^2 + r(l)^2 d\Omega^2.$$

In this form, the two sides correspond to

$$l > 0$$

and

$$l < 0.$$

Each side reaches a matching surface at

$$l = \pm l_a.$$

At these two surfaces, the interior geometry may be matched to exterior Schwarzschild regions with the same exterior mass parameter fixed by

$$M_{\text{out}} = \frac{c^2 R}{2G} \tilde{b}(x_a).$$

Taken together, these results indicate that the throat is not supported by an additional thin shell or singular matter layer placed at the throat itself. In the proper-distance coordinate, the throat appears simply as a regular point where the areal radius reaches its minimum. Furthermore, since the thin-shell-free condition is satisfied at the matching surfaces $r = a = Rx_a$ on both sides, the two interior branches may be smoothly connected to Schwarzschild exterior regions.

Thus, within the static and spherically symmetric setting considered here, the opened-throat region admits a natural two-sided extension in proper-distance coordinates. This provides a consistent static two-sided traversable-throat candidate, rather than a fully established dynamical wormhole solution. In particular, the time-dependent stability of this two-sided configuration remains to be analyzed separately.

XII. DISCUSSION

The analysis developed above shows that the minimal complex-conjugate scale selects a marginal throat through the pure geometric response. In the pure sector, the temporal equation gives $b'(R) = 1$, and therefore the

Morris–Thorne flare-out condition is saturated. Thus the characteristic scale $r = R$ is not an arbitrary throat radius, but the threshold point at which the pure response reaches the marginal throat state.

In the black-hole interpretation, the same characteristic scale can be identified with $R = GM/c^2$. At the level of the pure response, this scale corresponds to a marginal, non-opened throat. The condition is $b'(R) = 1$, not $b'(R) < 1$. Therefore, $R = GM/c^2$ should be understood as the location of a threshold throat selected by the pure geometric response. It is not, by itself, a fully opened throat.

The throat-opening tensor $\mathcal{Q}^\mu{}_\nu$ supplies the internal correction required to move the geometry away from this marginal state. Its temporal component lowers the throat derivative below unity, while its radial component is required to vanish at the throat in order to keep the redshift function finite. These two requirements are expressed by

$$\mathcal{Q}^t{}_t(R) < 0, \quad \mathcal{Q}^r{}_r(R) = 0.$$

The first condition controls the flare-out inequality, and the second controls the local regularity of the redshift function.

In the present construction, the throat-opening tensor is built from the variational residual window. The residual $\delta(r)$ measures the deviation of the scalar-field profile from the exact Euler–Lagrange extremum condition. After normalization, it defines the window function $W_\delta(x)$, which localizes the opening correction. With the representative choice $\alpha = \delta_{\text{max}}$, the throat derivative becomes

$$b'(R) = 1 - \delta(R_{(\sqrt{3})}) \simeq 0.615.$$

This satisfies

$$0 < b'(R) < 1,$$

so the marginal throat is locally opened at the level of the Morris–Thorne flare-out condition.

The radial sector provides an additional consistency structure. The radial matching coefficient β is fixed by the supply peak of the normalized exchange current, giving

$$\beta = \tilde{J}_{\text{sup}}^{\text{max}} \simeq 1.739.$$

With this value, the matching condition admits an inner solution at

$$x_a \simeq 2.416.$$

This solution is selected over the outer branch since the radial metric factor remains positive beyond the throat up to the matching surface. In particular, the zero of $D(x) = 1 - \tilde{b}(x)/x$ occurs at $x_h \simeq 3.217$, while the inner matching radius satisfies $x_a < x_h$. Thus, $1 < x \leq x_a$ has no horizon-like zero of the radial metric factor.

For the same inner branch, the redshift function remains finite up to the matching surface. The curvature

invariants evaluated in the throat region are also finite. In particular, the Ricci scalar, the Ricci tensor squared, and the Kretschmann scalar remain finite both near the throat and at $x_a \simeq 2.416$. Therefore, the resulting geometry can be interpreted as a locally nonsingular opened throat region, rather than merely as a formal flare-out condition.

It is useful to clarify how this opened sector would appear under an effective right-hand-side interpretation. Although the opening sector is treated here as a geometric contribution, one may formally move it to the right-hand side and read it as an effective stress-energy contribution:

$$\kappa T^\mu{}_{\nu, \text{eff}} = -\Lambda(r)\delta^\mu{}_\nu - Q^\mu{}_\nu.$$

With the mixed-component convention

$$T^\mu{}_\nu = \text{diag}(-\rho, p_r, p_t, p_t),$$

and with the explicit r -dependence suppressed, this gives

$$\rho_{\text{eff}} = \frac{\Lambda + Q^t_t}{\kappa}, \quad p_{r, \text{eff}} = -\frac{\Lambda + Q^r_r}{\kappa}, \quad p_{t, \text{eff}} = -\frac{\Lambda + Q^\theta_\theta}{\kappa}.$$

Therefore, the radial null combination is

$$\rho_{\text{eff}} + p_{r, \text{eff}} = \frac{Q^t_t - Q^r_r}{\kappa}.$$

At the throat, using the local opening conditions $Q^t_t(R) < 0$ and $Q^r_r(R) = 0$, one obtains

$$\rho_{\text{eff}}(R) + p_{r, \text{eff}}(R) < 0.$$

For this inner branch, we define the dimensionless diagnostics by

$$\bar{\rho} = \kappa R^2 \rho_{\text{eff}}, \quad \bar{p}_r = \kappa R^2 p_{r, \text{eff}}, \quad \bar{p}_t = \kappa R^2 p_{t, \text{eff}}.$$

Numerically, these quantities satisfy

$$\bar{\rho} > 0$$

throughout $1 \leq x \leq x_a$. The radial null combination obeys

$$\bar{\rho} + \bar{p}_r < 0$$

only in a near-throat region, crosses zero at $x \simeq 1.584$, located just before the recovery peak $x_{\text{rec}} \simeq 1.592$ ($\tilde{J}(x_{\text{rec}}) \simeq -0.443$), and remains non-negative up to the matching surface at $x_a \simeq 2.416$. By contrast, the transverse null combination, computed using the conservation-fixed Q^θ_θ , satisfies

$$\bar{\rho} + \bar{p}_t > 0$$

throughout the same interval. These diagnostics show that, under a formal right-hand-side interpretation, the opening sector can play the effective role usually assigned

to exotic matter support. However, in the present formulation, this sector is consistently retained on the geometric side of the field equations, since it is constructed from the internally generated geometric response rather than introduced as an independent matter source.

The exterior interpretation is more delicate. For the inner matching branch, the exterior Schwarzschild radius is determined by the internal shape function as

$$\frac{r_{s, \text{out}}}{R} = \tilde{b}(x_a).$$

With $\tilde{b}(x_a) \simeq 2.201$, this gives

$$r_{s, \text{out}} \simeq 2.201R, \quad a \simeq 2.416R.$$

The matching surface lies outside the exterior Schwarzschild radius. At the same time, the exterior photon-sphere radius is

$$\frac{r_{\text{ph}, \text{out}}}{R} = \frac{3}{2}\tilde{b}(x_a),$$

and therefore

$$r_{\text{ph}, \text{out}} \simeq 3.302R.$$

Hence,

$$r_{s, \text{out}} < a < r_{\text{ph}, \text{out}}.$$

The photon sphere therefore remains in the exterior Schwarzschild region. This means that the throat-opening construction does not immediately remove the usual photon-sphere-controlled shadow structure.

In terms of the internal scale R , the corresponding Schwarzschild shadow diameter is

$$\frac{D_{\text{sh}}}{R} = 3\sqrt{3}\tilde{b}(x_a).$$

This expression follows from the standard Schwarzschild photon-sphere and shadow-scale relation [49, 50]. For the accepted inner branch, this gives

$$\frac{D_{\text{sh}}}{R} \simeq 11.44.$$

This increase relative to the standard Schwarzschild value $6\sqrt{3} \simeq 10.39$ [50] reflects the fact that the exterior mass inferred from the matching is larger than the internal reference mass, with $M_{\text{out}} \simeq 1.10M$. This value is not used here as a direct observational fit, but it gives the characteristic shadow scale associated with the accepted matching branch. Its role is to show that the matching branch places the photon sphere outside the matching surface and remains compatible with a Schwarzschild-like exterior shadow region, a quantity of observational interest in horizon-scale imaging [51–54].

The distinction between a static two-sided throat candidate and a dynamically stable wormhole solution should be kept explicit. The analysis above supports the former, but not the latter. The remaining task is to determine whether this static two-sided configuration is dynamically stable under time-dependent perturbations, and whether its exterior matching leads to distinguishable wave-scattering or echo signatures [40–44].

XIII. CONCLUSION

In this paper, we studied the wormhole marginal throat selected by the minimal complex-conjugate scale $\zeta_{\pm} = x \pm i$, with $x = r/R$, and derived the local conditions required to open it. The complex-conjugate pair produces the real background response $\mathcal{B}(x) = \ln \zeta_+ + \ln \zeta_- = \ln(1 + x^2)$. From this response, the scalar profile $\phi(r) = 2r/(R^2 + r^2)$ is obtained, and the radius-dependent geometric response $\Lambda(r) = \phi(r)^2 = 4r^2/(R^2 + r^2)^2$ follows. Thus, $\Lambda(r)$ is not introduced as an arbitrary radial function, but is generated by the minimal complex-conjugate scale.

The response $\Lambda(r)$ reaches its maximum at the characteristic scale $r = R$, where $\Lambda(R) = 1/R^2$. For the Morris–Thorne-type geometry, the pure geometric sector $G^\mu{}_\nu + \Lambda(r)\delta^\mu{}_\nu = 0$ gives $b'(r) = r^2\Lambda(r)$ from the temporal component. Evaluating this relation at $r = R$, one obtains $b'(R) = R^2\Lambda(R) = 1$. Therefore, the pure geometric response selected by the minimal complex-conjugate scale does not directly produce an opened throat. Instead, it reaches the boundary value $b'(R) = 1$, saturating the flare-out condition as an equality, and forms a marginal throat.

To open this marginal throat, we introduced the throat-opening tensor $\mathcal{Q}^\mu{}_\nu$. With this tensor, the corrected geometric equation is written as $G^\mu{}_\nu + \Lambda(r)\delta^\mu{}_\nu + \mathcal{Q}^\mu{}_\nu = 0$. The local opening conditions at the throat are $\mathcal{Q}^t{}_t(R) < 0$ and $\mathcal{Q}^r{}_r(R) = 0$. The first condition lowers $b'(R)$ below unity and realizes the flare-out condition. The second condition prevents the redshift function from diverging at the throat.

The opening tensor was constructed using the variational residual of the scalar-field quasi-solution. Defining the normalized residual window by $W_\delta(x) = (\delta(Rx)/\delta_{\max})|_{R=R(\sqrt{3})}$, we used the temporal component $\mathcal{Q}^t{}_t(r) = -\alpha W_\delta(x)\Lambda(r)$, with $0 \leq \alpha \leq \delta_{\max}$. Then the throat derivative becomes $b'(R) = 1 - \alpha W_\delta(1)$. Choosing $\alpha = \delta_{\max}$, this reduces to $b'(R) = 1 - \delta(R(\sqrt{3}))$. Using the numerical value $\delta(R(\sqrt{3})) \simeq 0.385$, we obtain $b'(R) \simeq 0.615$. Thus, $0 < b'(R) < 1$ is satisfied, and the marginal throat is locally opened by the opening tensor constructed from the variational residual window.

The radial matching analysis further shows that the radial coefficient is fixed by the exchange-current supply peak. Explicitly, $\tilde{J}(x) = 8x(1 - x^2)/(1 + x^2)^3$, $x_{\text{sup}} \simeq 0.363$, and $\beta = \tilde{J}(x_{\text{sup}}) \simeq 1.739$. With this value, the regular inner matching branch appears at $x_a \simeq 2.416$. Along this branch, the radial metric factor $D(x)$ remains positive up to the matching surface, and the redshift function and curvature invariants remain finite. With the black-hole interior identification $R = GM/c^2$, the matching surface lies outside the exterior Schwarzschild radius and inside the exterior photon-sphere radius,

$$r_{s,\text{out}} < a < r_{\text{ph},\text{out}}, \quad 2.201R < 2.416R < 3.302R.$$

These results support the interpretation of the construction as a locally nonsingular opened-throat region with a possible Schwarzschild exterior connection.

The same inner branch also satisfies the smooth matching condition between the interior and exterior redshift derivatives,

$$\Phi'_{\text{in}}(x_a) = \frac{\tilde{b}(x_a)}{2x_a^2 D(x_a)} = \Phi'_{\text{out}}(x_a).$$

Thus, the accepted branch can be connected to a Schwarzschild exterior without introducing a thin shell at the matching surface, with the redshift gradients agreeing at the finite value $\simeq 2.126$.

Crucially, the accepted inner branch admits a natural two-sided extension in the proper-distance coordinate. In this description, the throat appears as a regular minimum of the areal radius, and the two sides correspond to the branches $l > 0$ and $l < 0$. In particular, at the throat,

$$\left. \frac{dr}{dl} \right|_{l=0} = 0, \quad \left. \frac{d^2 r}{dl^2} \right|_{l=0} > 0.$$

For the symmetric two-sided extension, the two exterior branches have identical matching radii, Schwarzschild radii, and exterior mass parameters, with $a_+ = a_- \simeq 2.416R$, $r_{s,+} = r_{s,-} \simeq 2.201R$, and $M_+ = M_- = M_{\text{out}}$. Thus, within the present static and spherically symmetric setting, the accepted branch gives a consistent two-sided traversable-throat candidate matched to Schwarzschild exterior regions on both sides.

In this construction, the minimal complex-conjugate scale first determines the real geometric response, and the pure part of that response selects the marginal throat. The variational residual then supplies the localization structure needed to build the opening tensor. In other words, the tension encoded in the residual between the kinetic-type and potential-type response components serves as an internal geometric window for opening the marginal throat.

The present work does not claim a fully established dynamical wormhole solution. The result established here is a static and spherically symmetric construction: the minimal complex-conjugate scale selects a marginal throat, the variational-residual-based opening tensor opens it, the exchange-current supply supports the exterior matching, and the proper-distance construction allows a static two-sided extension. A full dynamical stability analysis under time-dependent perturbations remains an important subject for future work. Such an analysis would first require formulating the (t, r) -dependent opening sector,

$$\Phi(t, r), \quad b(t, r), \quad \mathcal{Q}^\mu{}_\nu(t, r),$$

and then perturbing it around the static background to test whether the resulting linearized system is free of growing modes, with the causal bound $R(t) \geq GM/c^2$.

-
- [1] N. Dadhich, S. Kar, S. Mukherji, and M. Visser, $R = 0$ spacetimes and self-dual Lorentzian wormholes, *Phys. Rev. D* **65**, 064004 (2002), [arXiv:gr-qc/0109069 \[gr-qc\]](#).
 - [2] T. Damour and S. N. Solodukhin, Wormholes as black hole foils, *Phys. Rev. D* **76**, 024016 (2007), [arXiv:0704.2667 \[gr-qc\]](#).
 - [3] M. Cataldo, L. Liempi, and P. Rodríguez, Traversable Schwarzschild-like wormholes, *Eur. Phys. J. C* **77**, 748 (2017).
 - [4] S. Krasnikov, Schwarzschild-Like Wormholes as Accelerators, *Phys. Rev. D* **98**, 104048 (2018), [arXiv:1807.00890 \[gr-qc\]](#).
 - [5] P. O. Mazur and E. Mottola, Gravitational vacuum condensate stars, *Proc. Natl. Acad. Sci. U.S.A.* **101**, 9545 (2004), [arXiv:gr-qc/0407075 \[gr-qc\]](#).
 - [6] M. Visser and D. L. Wiltshire, Stable gravastars—an alternative to black holes?, *Class. Quantum Grav.* **21**, 1135 (2004), [arXiv:gr-qc/0310107 \[gr-qc\]](#).
 - [7] J. Bardeen, Non-Singular General-Relativistic Gravitational Collapse, in *Proceedings of the 5th International Conference on Gravitation and the Theory of Relativity* (Pub. House of Tbilisi University, Tbilisi, 1968) p. 174.
 - [8] I. Dymnikova, Vacuum nonsingular black hole, *Gen. Rel. Grav.* **24**, 235 (1992).
 - [9] S. A. Hayward, Formation and Evaporation of Nonsingular Black Holes, *Phys. Rev. Lett.* **96**, 031103 (2006), [arXiv:gr-qc/0506126 \[gr-qc\]](#).
 - [10] A. Simpson and M. Visser, Regular Black Holes with Asymptotically Minkowski Cores, *Universe* **6**, 8 (2020), [arXiv:1911.01020 \[gr-qc\]](#).
 - [11] A. Simpson and M. Visser, Black-bounce to traversable wormhole, *JCAP* **2019** (02), 042, [arXiv:1812.07114 \[gr-qc\]](#).
 - [12] F. S. N. Lobo, M. E. Rodrigues, M. V. d. S. Silva, A. Simpson, and M. Visser, Novel black-bounce spacetimes: Wormholes, regularity, energy conditions, and causal structure, *Phys. Rev. D* **103**, 084052 (2021), [arXiv:2009.12057 \[gr-qc\]](#).
 - [13] K. A. Bronnikov and R. K. Walia, Field sources for Simpson–Visser spacetimes, *Phys. Rev. D* **105**, 044039 (2022), [arXiv:2112.13198 \[gr-qc\]](#).
 - [14] F. D’Ambrosio and C. Rovelli, How Information Crosses Schwarzschild’s Central Singularity, *Class. Quantum Grav.* **35**, 215010 (2018), [arXiv:1803.05015 \[gr-qc\]](#).
 - [15] A. S. Agrawal, S. Zerbini, and B. Mishra, Black Holes and Wormholes beyond Classical General Relativity, *Phys. Dark Univ.* **46**, 101637 (2024), [arXiv:2406.01241 \[gr-qc\]](#).
 - [16] R. A. Konoplya and O. S. Stashko, Transition from Regular Black Holes to Wormholes in Covariant Effective Quantum Gravity: Scattering, Quasinormal Modes, and Hawking Radiation, *Phys. Rev. D* **111**, 084031 (2025), [arXiv:2502.05689 \[gr-qc\]](#).
 - [17] M. S. Morris and K. S. Thorne, Wormholes in spacetime and their use for interstellar travel: A tool for teaching general relativity, *Am. J. Phys.* **56**, 395 (1988).
 - [18] M. S. Morris, K. S. Thorne, and U. Yurtsever, Wormholes, Time Machines, and the Weak Energy Condition, *Phys. Rev. Lett.* **61**, 1446 (1988).
 - [19] M. Visser, *Lorentzian Wormholes: From Einstein to Hawking* (American Institute of Physics, Melville, NY, 1996).
 - [20] D. Hochberg and M. Visser, Geometric structure of the generic static traversable wormhole throat, *Phys. Rev. D* **56**, 4745 (1997), [arXiv:gr-qc/9704082 \[gr-qc\]](#).
 - [21] M. Visser and D. Hochberg, Geometric wormhole throats, *Ann. Isr. Phys. Soc.* **13**, 249 (1997), [arXiv:gr-qc/9710001 \[gr-qc\]](#).
 - [22] C. Barceló and M. Visser, Traversable wormholes from massless conformally coupled scalar fields, *Phys. Lett. B* **466**, 127 (1999), [arXiv:gr-qc/9908029 \[gr-qc\]](#).
 - [23] C. Barceló and M. Visser, Scalar fields, energy conditions and traversable wormholes, *Class. Quantum Grav.* **17**, 3843 (2000), [arXiv:gr-qc/0003025 \[gr-qc\]](#).
 - [24] F. S. N. Lobo, Exotic solutions in General Relativity: Traversable wormholes and “warp drive” spacetimes (2007), [arXiv:0710.4474 \[gr-qc\]](#).
 - [25] W. Israel, Singular hypersurfaces and thin shells in general relativity, *Nuovo Cim. B* **44**, 1 (1966).
 - [26] M. Visser, Traversable wormholes from surgically modified Schwarzschild spacetimes, *Nucl. Phys. B* **328**, 203 (1989), [arXiv:0809.0927 \[gr-qc\]](#).
 - [27] E. Poisson and M. Visser, Thin-shell wormholes: Linearization stability, *Phys. Rev. D* **52**, 7318 (1995), [arXiv:gr-qc/9506083 \[gr-qc\]](#).
 - [28] B. E. Taylor, W. A. Hiscock, and P. R. Anderson, Stress-energy of a quantized scalar field in static wormhole spacetimes, *Phys. Rev. D* **55**, 6116 (1997), [arXiv:gr-qc/9608036 \[gr-qc\]](#).
 - [29] D. Hochberg, A. A. Popov, and S. V. Sushkov, Self-Consistent Wormhole Solutions of Semiclassical Gravity, *Phys. Rev. Lett.* **78**, 2050 (1997), [arXiv:gr-qc/9701064 \[gr-qc\]](#).
 - [30] A. A. Popov, Stress-energy of a quantized scalar field in static wormhole spacetimes, *Phys. Rev. D* **64**, 104005 (2001), [arXiv:hep-th/0109166 \[hep-th\]](#).
 - [31] N. R. Khusnutdinov, Semiclassical wormholes, *Phys. Rev. D* **67**, 124020 (2003), [arXiv:hep-th/0304176 \[hep-th\]](#).
 - [32] V. B. Bezerra, E. R. Bezerra de Mello, N. R. Khusnutdinov, and S. V. Sushkov, Vacuum stress-energy tensor of a massive scalar field in a wormhole spacetime, *Phys. Rev. D* **81**, 084034 (2010), [arXiv:1003.0344 \[gr-qc\]](#).
 - [33] E. Kocuper, J. Matyjasek, and K. Zwierzchowska, Stress-energy tensor of quantized massive fields in static wormhole spacetimes, *Phys. Rev. D* **96**, 104057 (2017), [arXiv:1709.09034 \[gr-qc\]](#).
 - [34] J. Matyjasek, Stress-energy of the quantized fields in the spacetime of the Damour–Solodukhin wormhole, *Phys. Rev. D* **102**, 024082 (2020), [arXiv:2005.07077 \[gr-qc\]](#).
 - [35] A. R. Khabibullin, N. R. Khusnutdinov, and S. V. Sushkov, The Casimir effect in a wormhole spacetime, *Class. Quantum Grav.* **23**, 627 (2006), [arXiv:hep-th/0510232 \[hep-th\]](#).
 - [36] F. Sorge, Casimir effect around an Ellis wormhole, *Int. J. Mod. Phys. D* **29**, 2050002 (2020).
 - [37] A. C. L. Santos, C. R. Muniz, and L. T. Oliveira, Casimir Effect in a Schwarzschild-Like Wormhole Spacetime, *Int. J. Mod. Phys. D* **30**, 2150032 (2021), [arXiv:2007.00227 \[gr-qc\]](#).
 - [38] M. R. Mehdizadeh and A. H. Ziaie, Novel Casimir wormholes in Einstein gravity, *Eur. Phys. J. Plus* **139**, 1001 (2024), [arXiv:2406.03588 \[gr-qc\]](#).

- [39] V. Cardoso and P. Pani, Testing the nature of dark compact objects: a status report, *Living Rev. Rel.* **22**, 4 (2019), [arXiv:1904.05363 \[gr-qc\]](#).
- [40] V. Cardoso, S. Hopper, C. F. B. Macedo, C. Palenzuela, and P. Pani, Echoes of ECOs: gravitational-wave signatures of exotic compact objects and of quantum corrections at the horizon scale, *Phys. Rev. D* **94**, 084031 (2016), [arXiv:1608.08637 \[gr-qc\]](#).
- [41] P. Bueno, P. A. Cano, F. Goelen, T. Hertog, and B. Vercnocke, Echoes of Kerr-like wormholes, *Phys. Rev. D* **97**, 024040 (2018), [arXiv:1711.00391 \[gr-qc\]](#).
- [42] Y. Yang, D. Liu, Z. Xu, Y. Xing, S. Wu, and Z.-W. Long, Echoes of novel black-bounce spacetimes, *Phys. Rev. D* **104**, 104021 (2021), [arXiv:2107.06554 \[gr-qc\]](#).
- [43] S.-s. Bao, S. Hou, and H. Zhang, Searching for Wormholes with Gravitational Wave Scattering, *Eur. Phys. J. C* **83**, 127 (2023), [arXiv:2201.05866 \[gr-qc\]](#).
- [44] E. Franzin, S. Liberati, J. Mazza, R. Dey, and S. Chakraborty, Scalar perturbations around rotating regular black holes and wormholes: Quasinormal modes, ergoregion instability, and superradiance, *Phys. Rev. D* **105**, 124051 (2022), [arXiv:2201.01650 \[gr-qc\]](#).
- [45] M. Calzá, GrayHawk: A public code for calculating the Gray Body Factors of massless fields around spherically symmetric Black Holes, *Phys. Dark Univ.* **48**, 101900 (2025), [arXiv:2502.04041 \[gr-qc\]](#).
- [46] K. Schwarzschild, Über das Gravitationsfeld eines Massenpunktes nach der Einsteinschen Theorie, *Sitzungsber. Preuss. Akad. Wiss. Berlin*, 189 (1916).
- [47] A. Friedmann, Über die Krümmung des Raumes, *Z. Physik* **10**, 377 (1922).
- [48] S. Weinberg, *Gravitation and Cosmology: Principles and Applications of the General Theory of Relativity* (John Wiley & Sons, New York, 1972).
- [49] S. Chandrasekhar, *The mathematical theory of black holes*, The International series of monographs on physics, Vol. 69 (Clarendon Press, Oxford, 1983).
- [50] J.-P. Luminet, Image of a Spherical Black Hole with Thin Accretion Disk, *Astron. Astrophys.* **75**, 228 (1979).
- [51] Event Horizon Telescope Collaboration, First M87 Event Horizon Telescope Results. VI. The Shadow and Mass of the Central Black Hole, *Astrophys. J. Lett.* **875**, L6 (2019), [arXiv:1906.11243 \[astro-ph.GA\]](#).
- [52] Event Horizon Telescope Collaboration, The Persistent Shadow of the Supermassive Black Hole of M 87. I. Observations, Calibration, Imaging, and Analysis, *Astron. Astrophys.* **681**, A79 (2024).
- [53] Event Horizon Telescope Collaboration, First Sagittarius A* Event Horizon Telescope Results. IV. Variability, Morphology, and Black Hole Mass, *Astrophys. J. Lett.* **930**, L15 (2022), [arXiv:2311.08697 \[astro-ph.HE\]](#).
- [54] S. Vagnozzi, R. Roy, Y.-D. Tsai, L. Visinelli, M. Afrin, A. Allahyari, P. Bambhaniya, D. Dey, S. G. Ghosh, P. S. Joshi, K. Jusufi, M. Khodadi, R. Kumar Walia, A. Övgün, and C. Bambi, Horizon-scale tests of gravity theories and fundamental physics from the Event Horizon Telescope image of Sagittarius A*, *Class. Quantum Grav.* **40**, 165007 (2023), [arXiv:2205.07787 \[gr-qc\]](#).